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Modeling flexible behavior with remapping-based hippocampal sequence learning

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eLife Assessment

This manuscript reports a potentially **valuable** modeling study on sequence generation in the hippocampus in a variety of behavioral contexts. While the scope of the model is ambitious, its presentation is **incomplete**, and there remains some lack of clarity on the methodology and interpretation. The work will interest the broad community of researchers studying cortical-hippocampal interactions and sequences.

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Abstract

Animals flexibly change their behavior depending on context. It is reported that the hippocampus is one of the most prominent regions for contextual behaviors, and its sequential activity shows context dependency. However, how such context-dependent sequential activity is established through reorganization of neuronal activity (remapping) is unclear. To better understand the formation of hippocampal activity and its contribution to context-dependent flexible behavior, we present a novel biologically plausible reinforcement learning model. In this model, Context selector promotes the formation of context-dependent sequential activity and allows for flexible switching of behavior in multiple contexts. This model reproduces a variety of findings from neural activity, optogenetic inactivation, human fMRI, and clinical research. Furthermore, our model predicts that imbalances in the ratio between sensory and contextual representations in Context selector account for schizophrenia (SZ) and autism spectrum disorder (ASD)-like behaviors.

Introduction

Humans exhibit highly flexible behavior. However, a major challenge in solving various tasks with one neural network is that the same external stimulus can have different meanings depending on the context. For example, the word "mouse" can mean either an animal or a PC device, depending on the context (Figure 1A). Therefore, for correct word recognition, the biological neural computation should not be based only on the word "mouse" alone, but also on the context it appears in. In experiments, it is reported that the hippocampus is one of the most important regions for contextual behavior. Hippocampal neurons show sequential activity (Buzsáki and Tingley, 2018; Skaggs and McNaughton, 1996; Wilson and McNaughton, 1993) related to episodic memory (Burgess et al., 2002), the amount of reward (Ambrose et al., 2016), planning (Ólafsdóttir et al., 2018), and recall (Carr et al., 2011), and their representation depends on the context (Hasselmo and Eichenbaum, 2005). Additionally, hippocampal neurons exhibit reorganized neural activity called remapping (Bostock et al., 1991; Muller and Kubie, 1987), which does not purely reflect the change in the external stimuli but task structure (Jeffery et al., 2003), and subjective context (Sanders et al., 2020). However, how context-dependent sequential activity in the hippocampus is established through remapping and how it contributes to flexible behavior remain to be understood.

Several theoretical models have been proposed to explain how hippocampal activity depends on context. The first approach uses the structure of the environment. The Tolman-Eichenbaum Machine (Whittington et al., 2020) and the Clone Structured Cognitive Graph (George et al., 2021) account for context-dependent neural activities, such as splitter cells (Dudchenko and Wood, 2014) and lap cells (Sun et al., 2020), by introducing graphical structure stored within the network. However, these models entail optimization procedures like backpropagation or the expectation-maximization (EM) algorithm (Whittington et al., 2020, George et al., 2021), which are not considered biologically plausible. The second approach uses eligibility trace to explain how past experiences, i.e., temporal context, are integrated into hippocampal activity (Cone and Clopath, 2024). In this framework, the length of the temporal context is constrained by the time constant of the eligibility trace.

Nevertheless, animals can flexibly estimate the current context using history of various lengths (Barnett et al., 2014), suggesting that hippocampal activity may not be bound by a fixed eligibility window. The third approach trains recurrent neural networks (RNNs) to replicate the dynamics of hippocampal activity (Leibold, 2020). While previous works have explored hippocampal sequential activity for planning (Jensen et al., 2024; Mattar and Daw, 2018; Pettersen et al., 2024; Stachenfeld et al., 2017) and hippocampal remapping for contextual inference (Low et al., 2023) separately, they have yet to elucidate how these two aspects jointly enable flexible behavior. A simple biologically plausible model-based reinforcement learning model that uses the Amari-Hopfield model for context selection and hippocampal sequences of various lengths as a state-transition model for long-horizon planning, relying on remapping driven by prediction errors to form state representation, would thus provide valuable insights into the neural mechanisms underpinning context-dependent flexible behavior.

We aim to understand how hippocampal remapping, driven by prediction errors, gives rise to the formation and use of context-dependent hippocampal sequences, providing a biologically plausible account of flexible behavior, including rodents and humans. Our key idea is as follows. When the external environment deviates from the expectations of the current subjective context, prediction errors arise and trigger remapping. This process recruits distinct subsets of neurons to encode novel experience, thereby establishing separate contextual memories and enabling flexible goal-oriented behavior in response to sudden environmental changes. To demonstrate the capability of this idea, we constructed a computational model comprising two modules: Context selector that selects the appropriate context based on prediction errors, and Sequence composer (hippocampus) that learns to compose neural activity sequences predicting future events by concatenating context-dependent hippocampal segments according to reward. Our model implements simple model-based reinforcement learning in ambiguous contexts, yielding flexible behavior using a biologically plausible synaptic plasticity rule. We show that it reproduces a range of context-dependent hippocampal activities as well as the impairments associated with specific brain lesion studies.

Finally, our model predicts a relationship between deficits in model-based behavior and sensory processing. Clinical research has reported that patients with schizophrenia (SZ) or autism spectrum disorder (ASD) often exhibit problems with both behavioral flexibility and sensory processing, including hyper- and hyposensitivity (Javitt and Freedman, 2015; Watts et al., 2016). These symptoms frequently co-occur, but the underlying reason remains unclear. Our model shows that the relative sizes of the neural populations in the sensory-processing region and the context-processing region within Context selector are important for contextual inference, suggesting that treatments targeting sensory processing could improve cognitive flexibility in some psychoses.

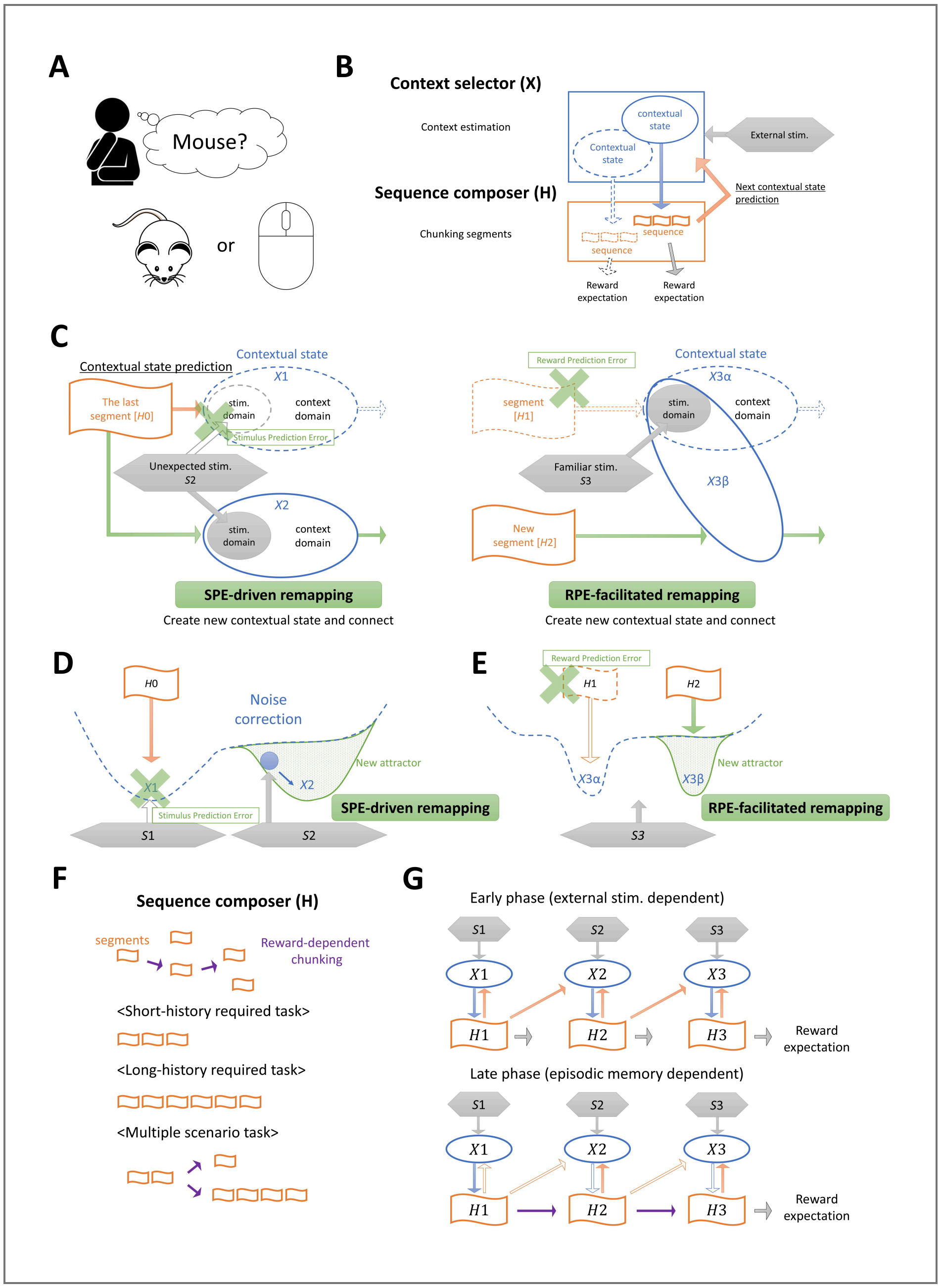

Figure 1. Schematic representation of our model.

A, An example of context-dependent cognition. Humans can understand the meaning of "mouse" (an animal or a computer input device) depending on the context. **B**, Our model involves two modules: Context selector (X) and Sequence composer (H). X chooses a context depending on the external stimuli and the input from H, and activates a sequence in H. This sequence is used for reward prediction. In addition, H sends predictive feedback about external stimuli to X. **C**, The schematic figure of two kinds of remapping. Grey boxes indicate external stimuli, orange boxes indicate hippocampal segment (a part of hippocampal sequence), blue circles indicate contextual state, and green cross marks indicate the prediction error about external stimuli (left) and about reward (right). Solid lines indicate the actual state transition and dotted lines indicate virtual state transition that is created in the past transition. Green arrows indicate the synaptic potentiation related to remapping. **D, E**, Attractor dynamics of Amari-Hopfield network related to SPE-driven remapping (**D**) and RPE-facilitated remapping (**E**). Blue dotted lines indicate an energy landscape, and green solid lines indicate the chosen attractor as a result of remapping. **F**, Hippocampal segments in H are combined depending on rewards (purple arrows) and formed into task-dependent sequences. Each sequence supports action planning and enables predictions of future external stimuli and rewards. **G,** An example state transition related to hippocampal sequence formation. In early phase, hippocampal neurons are activated through the input from X, while in the late phase, hippocampal neurons are activated through the recurrent input within H.

Results

As illustrated in Figure 1B, we modeled the neural mechanisms of context-dependent behavior as the interaction between two functional modules: Context selector (X), which selects appropriate contexts, and Sequence composer (hippocampus, H), which generates neural activity sequences that predict future events. We use the Amari-Hopfield network (Amari, 1972; Hopfield, 1982) with Hebbian plasticity for X. X has two domains: a stimulus domain that represents external stimuli, and a contextual domain that represents subjective contextual information. While the stimulus domain represents environmental states specified by the external stimuli, the contextual domain represents the contextual states for a given environmental states, which correspond to different subjective interpretations or associations of the external stimulus. X can stably store multiple contextual states by creating attractors in Amari-Hopfield model.

Our model's operations are algorithmic in nature indicated in Figure S1. When agents are at a starting point (i.e., a landmark), X initializes the neural activity of the contextual domain based on the external stimulus (see Materials and methods). When agents move to other environmental states, X receives predictive input from the lastly activated hippocampal segment together with the external stimulus and estimates the current context. Once X's contextual state is set, it transmits the resulting output to H, which then activates an initial segment of H's episodic sequence. H produces an episodic sequence corresponding to hippocampal replay (Davidson et al., 2009) or planning (Ólafsdóttir et al., 2018) based on its connectivity. For simplicity, we use a binary recurrent neural network for H, whose connectivity is updated by a three-factor Hebbian plasticity rule that depends on reward (see Materials and methods). Each replayed sequence is associated with actions (i.e. transition to the next environmental states) and two predictive outcomes: predicted future external stimuli and expected reward value. Based on the source of prediction errors, we consider two types of remapping: sensory prediction error (SPE)-driven remapping and reward prediction error (RPE)– facilitated remapping (Figure 1C). SPE-driven remapping is triggered when the mismatch between the predictive inputs from H to X and externally driven sensory inputs exceeds a threshold (see Materials and methods), causing X to either transition to a different contextual state or form a new one (Figure 1D). RPE-facilitated remapping is more likely to be triggered when the agents execute an action plan following a hippocampal sequence marked by a no-good indicator. The no-good indicator indicates that the action plan, i.e. the hippocampal sequence, has recently been associated with negative reward prediction errors, possibly due to environmental changes (see Materials and methods). It then facilitates the exploration of alternative hippocampal sequences (Figure 1E). At the beginning of learning, hippocampal segments are not connected, and H yields only short sequences that generate immediate actions and short-term predictions. As learning continues, the three-factor Hebbian plasticity rule concatenates these segments, thereby creating longer sequences that reflect the task structure (Figure 1F). Thus, H learns to generate extended sequences that outline a course of actions and predict both reward and subsequent changes in the environment without explicit inputs from X (Figure 1G), forming a simple transition model for model-based reinforcement learning (Coulom, 2007). If a significant reward prediction error arises from a sequence, the agent explores a random action not specified by that sequence (see Materials and methods).

In the framework of reinforcement learning, our model can be mapped onto a Bayesian-adaptive model-based architecture in which contextual state serves as the root of Monte Carlo tree search (Guez et al., 2013) in a simple, largely stable environment with noiseless and unambiguous sensory stimuli, and only occasional abrupt changes. In this setup, prediction errors arise from the agent's lack of experience or due to abrupt environmental changes. Once a context selector X infer the hidden state, the sequence composer H generates episodic sequences that correspond to trajectories in a search tree, each branch representing possible action–outcome sequences. Just as Monte Carlo tree search explores potential future paths to evaluate expected rewards, H produces hippocampal sequences that simulate future states and rewards based on its learned connectivity. In this way, X defines the context that anchors the root of the tree, while H expands the tree through replay or planning, thereby our model provides a simplified algorithmic implementation model-based reinforcement learning via tree search planning. However, these conceptual similarities are qualitative rather than quantitative. The goal of this work is not to achieve Bayesian optimality, but rather to show qualitative remapping-related processes that support goal-directed planning following epistemic errors.

Splitter cells

Our model reproduces a range of hippocampal activity patterns that align with empirical data. First, we confirmed that our model reproduces the splitter cells reported in the hippocampus (Dudchenko and Wood, 2014). Splitter cells are a subset of hippocampal neurons that fire differentially on an overlapping segment of trajectories depending on where the animal came from, and/or where it is going. It is known that they do so based on information that is not present in sensory or motor patterns at the time of the splitting effect, but rather appear to reflect the recent past, upcoming future, and/or inferences about the state of the environment (Duvelle et al., 2023).

Experimentally, splitter cells are most often observed in an alternation task in a modified T-maze. Here, we simplified this task by using an environment with five discrete states (S1 – S5), i.e. five discrete external stimuli (Figure 2A). In this environment, agents successfully solve this task by SPE-driven remapping, which creates different contextual states $X2\alpha$ and $X2\beta$ at an environmental state $S2$ based on where the agents came from, and thereby enabling context-specific exploration of which state to go ($S3$ or $S4$) (Figure 2B).

Figure 2C illustrates an example of both the environmental state transition and the corresponding contextual state transition of an agent. The neural activity of X at each contextual state is shown in Figure 2D, where the environmental states (e.g., $S1$, $S2$...) are represented in the stimulus domain and the contextual states (e.g., $X1$, $X2\alpha$...) are represented in the context domain. A second contextual state at $S2$, $X2\beta$, was generated through SPE-driven remapping at the second visit of S2 (second trial) due to history mismatch between $S1 \rightarrow S2$ ($X1 \rightarrow X2\alpha$) and $S3 \rightarrow S2$ ($X3 \rightarrow X2\beta$) (see Figure S1). In Sequence composer, two types of neurons exist: state-coding neurons, which represent each contextual state, and transition-coding neurons, which encode transitions to successive contextual states given the contextual state indicated by the state-coding neuron (see Materials and methods). Note that in the real brain, not only hippocampus but also the premotor cortex and the basal ganglia contribute to action planning and execution (Hikosaka et al., 2002). Here, however, we focus on how simplified planning sequences are learned and composed in a context-dependent manner. In the example transition shown in Figure 2C, the agent selected an environmental state transition from S2 to S4 in the 2$^{\text{nd}}$, 5$^{\text{th}}$, and 8$^{\text{th}}$ trials, which corresponds to a contextual state transition from $X2\beta$ to $X4\beta$ in the X module. However, because this transition was not rewarded, no synaptic potentiation occurred among hippocampal neurons. Subsequently, in the 11$^{\text{th}}$ trial, the agent attempted an environmental state transition from S2 to S5, which corresponds to the transition from $X2\beta$ to $X5\beta$ in the contextual states. The agent received a reward at S5, and the corresponding hippocampal sequence was strengthened, enabling the agent to acquire the alternation task in the following trials (Figure 2E).

In our model, most agents can solve this task (Figure 2F). As learning progresses, the length of hippocampal sequences increases, and eventually planning of the transition from one reward state to the next is possible (Figure 2G). Our model can be compared to the neural activity of the rats' splitter cells in the hippocampus during the modified T-maze task (Wood et al., 2000) (Figure 2H). In our model, the transition-coding neurons exhibit right/left turn–specific firing at S2 after learning is complete (Figure 2E, I), replicating the emergence of splitter cells.

Lap cells

The emergence of splitter cells explored above has also been studied in previous work (Duvelle et al., 2023; Hasselmo and Eichenbaum, 2005; Katz et al., 2007). However, these approaches generally assume that an appropriate temporal context—or a fixed length of sensory histories—is prepared in advance. This assumption becomes problematic in tasks where the number of required histories is unknown or changes dynamically: preparing too few histories results in failing to solve the tasks, while preparing too many slows down the search for a solution. Instead

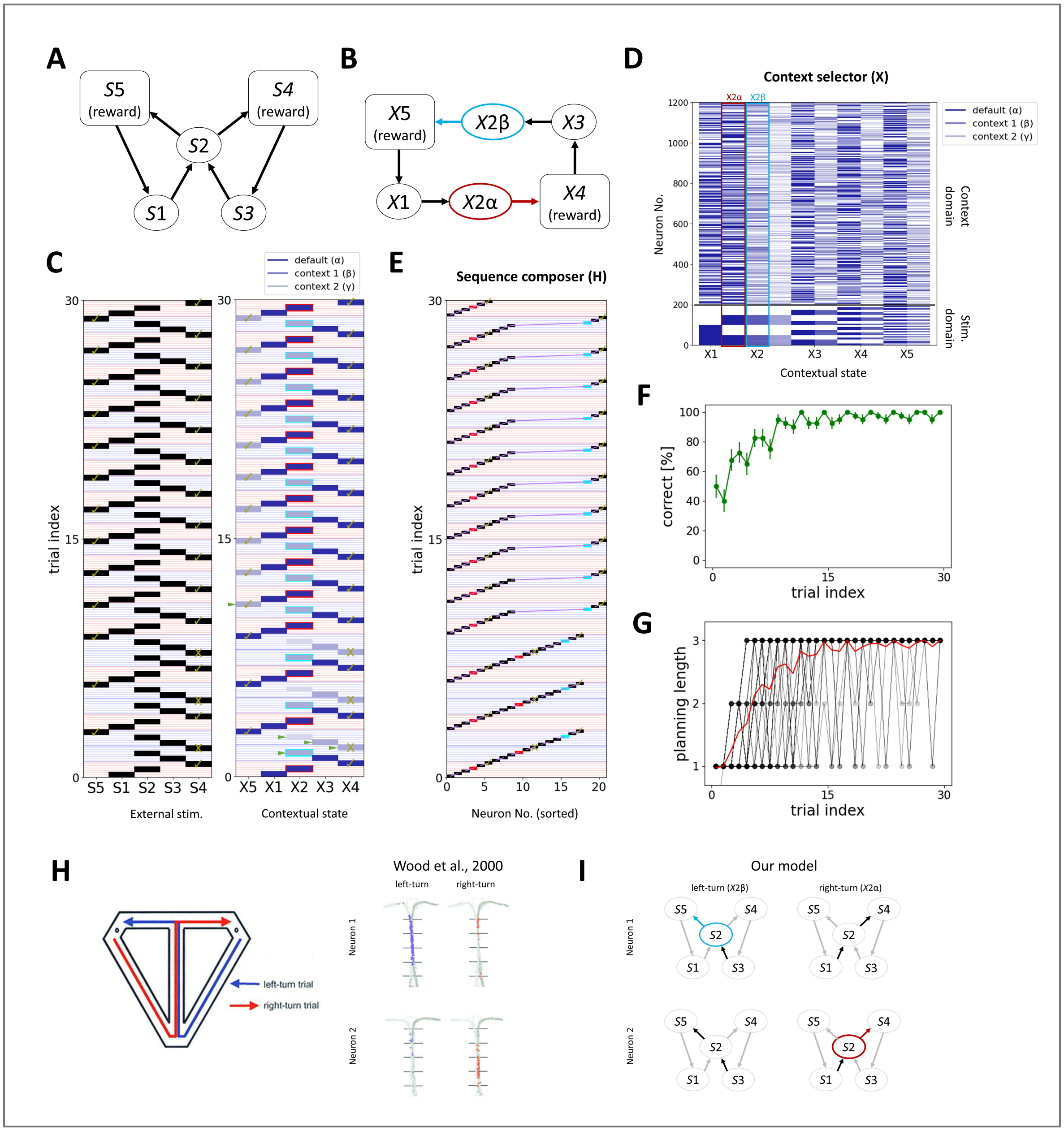

Figure 2. Our model replicates the emergence of splitter cells.

A, Simplified alternation task diagram. **B**, A successful contextual state transition of our model. Preparing 2 different contextual states *X2α* and *X2β* at S2 is necessary to solve this task. **C**, An example environmental state transition (left) and contextual state transition (right). Check marks indicate the rewarded states, and cross marks indicate non-rewarded states. Red shades indicate the right-turn trials and blue shades indicate the left-turn trials. (Right) The intensity of blue indicates the order of created contextual state, following history-driven remapping indicated in green triangles. Red outlines indicate *X2α* and blue outlines indicate *X2β*. **D**, The corresponding neural activity of X to each contextual state. The neurons in the stim. domain are sorted according to external stimuli. **E**, The corresponding hippocampal activity at each contextual state. Red square indicates the transition-coding neuron of S2 to S4, and blue square indicates the transition-coding neuron of S2 to S5. Purple line indicates the hippocampal sequence, which is gradually lengthened in reward-dependent manner. **F**, The correct rate of our model. The error bar indicates the standard error of the mean (N = 40). **G**, The maximum number of environmental states ahead that the agents planned (planning length) gradually increases over learning. Black lines indicate the planning length of each agent, and the red line is their average. **H**, Emergence of splitter cells in the hippocampus in the modified T-maze modification task (Wood et al., 2000). **I**, The transition-coding neurons in our model replicate the emergence of splitter cells in S2.

of preparing temporal context of fixed length in advance, our model uses remapping that adds new contextual states whenever a prediction error arises. This approach enables on-demand creation of contextual states and accelerates solution-finding in dynamically changing tasks.

To show the advantage of our model, we demonstrate that our model replicates the emergence of lap cells (Sun et al., 2020). We set up a simplified discrete environment with a loop structure where the number of laps required to receive a reward varies (Figure 3A). Agents are initially rewarded for the shortest transitions through environmental states $S1 \rightarrow S2 \rightarrow S4$. After 20 trials, the environment changes, and the agents are rewarded for one lap transition, i.e., $S1 \rightarrow S2 \rightarrow S3 \rightarrow S2 \rightarrow S4$. It causes a large reward prediction error (*no-good* indicator, see Materials and methods) and triggers RPE-facilitated remapping and exploration in the environment. During exploration, history mismatch triggers SPE-driven remapping in $S2$ and $S4$ as we showed in Figure 2, and contextual states are discriminated into $X2\alpha$ / $X2\beta$ and $X4\alpha$ / $X4\beta$ based on the history (i.e. laps). In Sequence Composer, the transition of contextual state $X1 \rightarrow X2\alpha \rightarrow X3\alpha \rightarrow X2\beta \rightarrow X4\beta$ is reinforced. After another 20 trials, the task environment changes again and the agents are rewarded for two laps, i.e., $S1 \rightarrow S2 \rightarrow S3 \rightarrow S2 \rightarrow S2 \rightarrow S3 \rightarrow S4$, or more. Either the shortest transition, $X1 \rightarrow X2\alpha \rightarrow X4\alpha$, or the one lap transition, $X1 \rightarrow X2\alpha \rightarrow X3\alpha \rightarrow X2\beta \rightarrow X4\beta$, is no longer rewarded, which triggers another RPE-facilitated remapping and exploration. During exploration, history mismatch occurs in $S2$, $S3$ and $S4$, and the contextual states for the second lap ($X2\gamma$, $X4\gamma$) are generated. Finally, the rewarded transition of contextual states and corresponding sequence, i.e., $X1 \rightarrow X2\alpha \rightarrow X3\alpha \rightarrow X2\beta \rightarrow X3\beta \rightarrow X2\gamma \rightarrow X4\gamma$, is reinforced (Figure 3B).

In our model, most agents can solve this task (Figure 3C). The episodic memory used for planning changes successfully depending on the environment (Figure 3D). This task is comparable with the 4- laps task for rats (Sun et al., 2020). In an environment where rats are rewarded for every four laps of a circuit, different hippocampal neurons fire for each lap. Our model replicates this result with the different hippocampal cells firing for different laps (Figure 3E). It is also reported that the inhibition of medial entorhinal cortex axons at CA1 attenuates the lap-specific activity (i.e., event-specific rate remapping (ESR)) without much affecting spatial encoding. Our model replicates this result by blocking the synaptic transmission from most ofneurons in the context domain of X to H (Figure 3F).

This task can also be solved by simply preparing temporal contexts with three steps of sensory history (n = 3), which is the minimal number to solve this task (see Materials and methods for Model-free learning). However, it takes much longer to find the correct transition for solving the 1-lap task than our model because it involves an excessive number of states (Figure S2). This result indicates that our model, which creates contextual states on demand, can perform better than the model with a fixed-length history.

To demonstrate the advantage of our model in a rapidly switching task that requires different history lengths, we show that an agent trained on both the 1-lap and 2-laps tasks can flexibly alternate between them in a reward-dependent manner (Figure 3G), selectively engaging hippocampal sequences of different lengths according to the current task context (Figure 3H). Together, these results illustrate how hippocampal lap-like representations emerge through learning and enable flexible context switching across tasks with distinct temporal demands.

Planning in a stimulus-cued dynamic environment

In the real world, external stimuli dynamically change, and animals make plans and derive appropriate behavior by using the external stimulus as a clue. Here, we demonstrate that our model replicates key features of stimulus-related contextual behavior and its neural activity reported in experimental studies using SPE-driven remapping.

We consider a simplified environment of probabilistic cueing paradigm (Ekman et al., 2022). In this study, two auditory contextual cues probabilistically predicted distinct visual motion sequences, and fMRI decoding was used to examine the frequency of hippocampal replay. We simplified this task as shown in Figure 4A. In initial environment I, agents start from S0 and go to a state where one of two different external stimuli $S2$ or $S3$ is presented with different

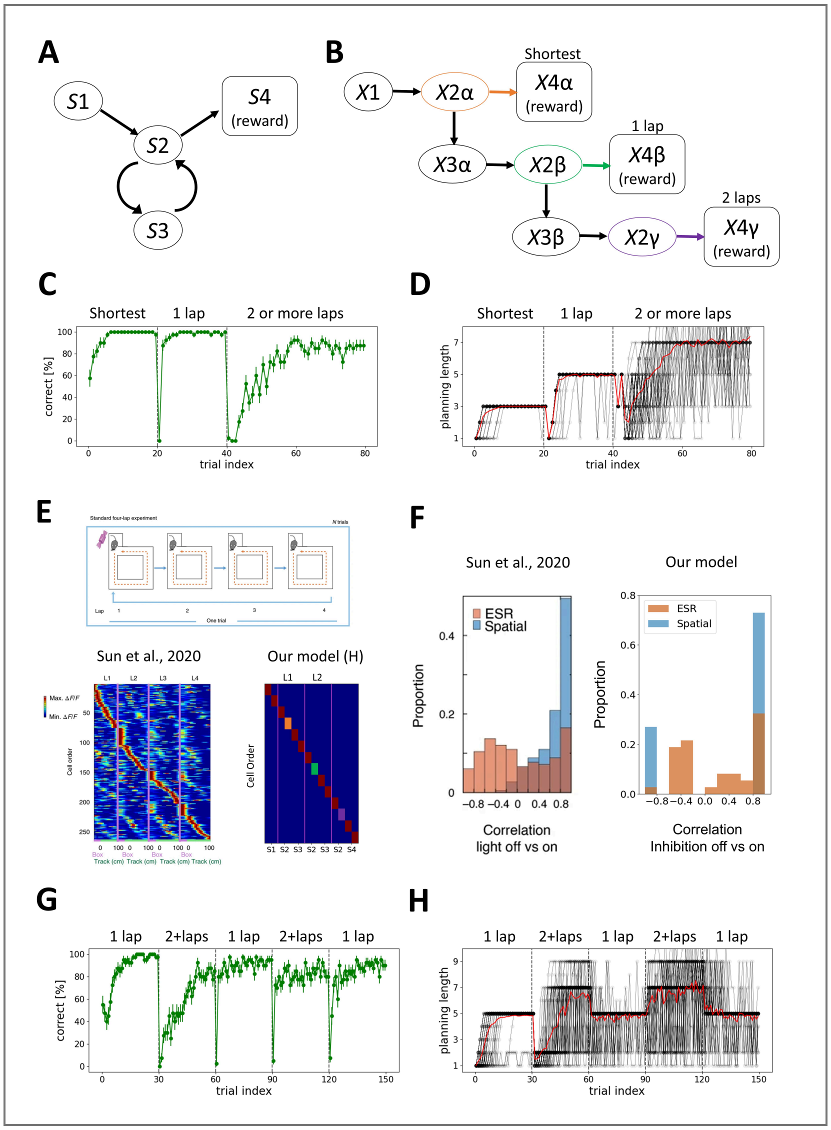

Figure 3. **Our model replicates the emergence of lap cells.**

A, Simplified 2-laps task diagram. Agents are rewarded for the shortest path (S1→S2→S4) for the initial 20 trials, for the 1-lap path (S1→S2→S3→S2→S4) for the next 20 trials, and for the 2 or more laps (S1→S2→S3→S2→S3→S2→S4, etc.) for the next 40 trials. **B**, A successful contextual state transition map of our model. The environmental states S2 and S4 are split into three contextual states *(X2α, X2β, X2γ),* S3 is split into two contextual states *(X3α, X3β),* and S4 is split into three contextual states (X4α, X4β, X4γ). **C**, The correct rate of our model. The error bar indicates the standard error of the mean (N = 40). **D**, The planning length gradually increases during learning, depending on the task demand. The black lines indicate the planning length of each agent, and the red line is their average. **E**, The comparison of (Left) lap cells in the hippocampus in the 4-laps task (Sun et al., 2020) and (Right) our results of active neurons in H module. The transition-coding neurons at S2 in 2-laps task are indicated in orange and green and purple squares corresponding to **B**. **F**, The inhibition experiment of medial entorhinal cortex axons at CA1. ESR cells show a weak lap-specific correlation (ESR correlation) between light-on trials and light-off trials, while they show a strong spatial correlation between light-on trials and light-off trials (Left). Our model replicates the result qualitatively with the inhibition on and off (Right). **G**, The correct rate of 1-lap and 2-or-more-laps alternation task. The error bar indicates the standard error of the mean (N = 40). **H**, The planning length adapts flexibly to the task demand.

probability (p=0.8, 0.2 respectively). When *S2* is presented, agents can get a reward at *S4*, whereas when S3 is presented, they can get a reward at *S5*. After 30 trials, the environment changes to II and the initial stimulus is switched to *S1*, not *S0*. In this environment, agents are rewarded at *S5* and *S4* when the external stimulus is *S2* and *S3*, respectively (i.e., Reversal).

In such a stochastic environment, the agents need to switch transition rules according to the external stimuli regardless of the prediction about the external stimuli beforehand. SPE-driven remapping (Figure 1D) enables our model to quickly change or generate the different context when the prediction error about the external stimuli occurs. For instance, in environment I, two rewarded contextual transitions exist: a more likely one ($X0 \rightarrow X2\alpha \rightarrow X4\alpha$) and a less likely one ($X0 \rightarrow X3\alpha \rightarrow X5\beta$) (Figure 4B). When an agent predict the major stimuli (*S*2) at the initial state (*S*0) but minor stimuli (*S*3) is presented, the agent stops the sequence-based action loop (Figure S1), and SPE-driven remapping occurs, which switches the contextual state from $X2\alpha$ to $X3\alpha$ and the corresponding hippocampal sequence. As a result, the agents choose the correct transition regardless of prior prediction (Figure 4B).

In our model, most agents can learn to make appropriate transitions depending on the external stimuli. Importantly, they show a one-shot switch between the environment I and II when the agents experience the environment for the second time (Figure 4C). This is because contextual states for *S*2 and *S*3 are generated differently for the environment I and II, i.e. $X2\alpha$, $X3\alpha$ for environment I and $X2\beta$, $X3\beta$ for environment II, through SPE-driven remapping. The length of the planning sequence used in the actual transition converges to between 2 and 3 because agents reselect the hippocampal sequence and the contextual state when the external stimuli differ from predictions and SPE-driven remapping is triggered (Figure 4D). The probability of predicted external stimuli (S2 or S3) based on the generated sequences matches with the actual probability (p = 0.8, 0.2, respectively) (Figure 4E), because of the reward-dependent synaptic plasticity in hippocampus (see Materials and methods). This result replicates Ekman et al. (2022), who showed that the probability of the contextual cues is reflected in the statistically significant differences in hippocampal replay probability in humans (Figure 4F).

Our model is applicable to context selection under ambiguous external stimuli. Julian and Doeller (2021) used a similar task structure as Figure 4A in humans and reported that the contextual representations and realignment in hippocampus under ambiguous external stimuli predict context-dependent behavior. In training phase, agents are put into either Square (Sq) or Circle (Ci) virtual reality arena, and then one of two target objects (S2 or S3) is randomly specified with equal probability. Depending on the arena type, the agents decide to transit to S4 or S5 to get reward. In test phase, subjects are put into either Sq, Ci, or their morphed version, Squircle (SC) arena, i.e. mean value of Sq and Ci. Under SC arena, the agents transit depending on the subjective context of either Sq or Ci. Note that reward feedback is not given in the test phase (Figure 4G).

Our model successfully learns this task, and the agents show context-dependent behaviors under Sq or Ci arena in the test phase (Figure 4H). Additionally, our model replicates the experimental results of SC as the mixed Sq- or Ci-like behaviors (Figure 4I). In humans, the Sq- or Ci-like behaviors are well decoded in hippocampus, but it degrades under SC condition (Julian and Doeller, 2021). Our model replicates this result with degraded decoding score under SC condition (Figure 4J). Here, three reconstruction cases are observed in X under SC condition: Sq context reconstruction, Ci context reconstruction, and a default context usage of SC due to X's failure to convergence (see Materials and methods). In the last case, the agents make a random transition by recruiting new hippocampal neurons. Therefore, behavioral decoding based on hippocampal neural activity is lower than that under the Sq and Ci conditions (Figure 4J). This result is consistent with the findings of Julian and Doeller (2021).

Prediction related to sensory processing and flexible behavior

Our model does not only replicate a variety of experimental results but also make predictions. In clinical research, it has been reported that issues related to behavioral flexibility and sensory processing often co-occur in certain psychiatric conditions, including schizophrenia (SZ) (Javitt and Freedman, 2015) and autism spectrum disorder (ASD) (Watts et al., 2016). Many studies have

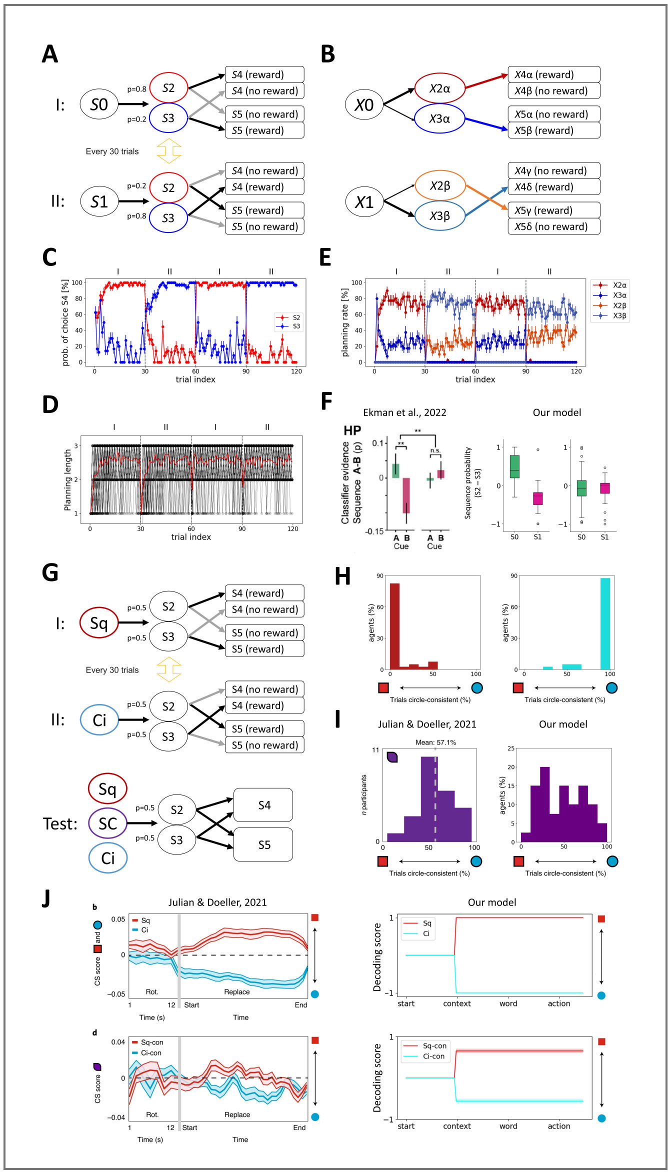

Figure 4. Our model replicates key features of human neural activity in dynamic environments.

A, Simplified probabilistic cueing task diagram. In environment I, agents start at S0 and move to S2 or S3 randomly (S2 for p = 0.8 and S3 for p = 0.2) and receive a reward in S4 when they come from S2 and in S5 otherwise. In environment II, agents start at S1 and move to S2 or S3 randomly (S2 for p = 0.2 and S3 for p = 0.8) and receive a reward in S5 when they come from S2 and in S4 otherwise. The environment switches between the two every 30 trials. **B**, A successful context map of this task. S2 and S3 are split into two contextual states, and S4 and S5 are split into four contextual states. The hippocampal connections are built for rewarded conditions only. **C**, The probability of choosing S4. The red/blue line shows its mean when S2/S3 is presented. The error bar indicates the standard error of the mean (N = 40). **D**, The planning length gradually increases over learning and converges to 3. The black lines indicate each agent's planning length, and the red line is their average. **E**, The probability of generating a specific planning sequence at S0 or S1. The expected states (S2 or S3) are modulated according to the environment. **F**, Our model behavior is similar to the human fMRI result of the cue-probability-dependent hippocampal replay (Ekman et al., 2022). Paired sample t-test. **P<0.01. **G**, Simplified task diagram (Julian and Doeller, 2021). The training phase is the same as **A**, but the contextual stimuli of Square (Sq) or Circle (Ci) are initially presented and the probability of S2 and S3 is equal. In the test phase, either one of Sq, Ci or the mixture stimuli of Sq and Ci (Squircle: SC) are presented, and the agent transfers following their faith. Reward feedback is not given in the test phase. **H**, The transition probability under Sq context (Left) and Ci context (Right). **I**, The transition probability under SC context of the human patients in Julian and Doeller, 2021 (Left) and our model (Right). **J**, Comparison of behavioral decoding accuracy from hippocampal fMRI activity of Julian and Doeller, 2021 (Left) and hippocampal neural activity of our model (Right). Our model replicates the worse decoding accuracy in SC context (Bottom) than Sq or Ci context (Top).

reported that both symptoms are linked to the dysfunction of the prefrontal cortex (PFC) (Kaplan et al., 2016; Watanabe et al., 2012); however, the reasons for their cooccurrence are not yet fully understood.

We assume that this dysfunction corresponds to hypo-/hyper-representation of stimulus information in X. To investigate this hypothesis, we altered the ratio of neurons in the context domain and sensory domain in X in our model. We used the same task described in Figure 4A with equal probability transitions to S2 and S3 (Figure 5A). When the stimulus domain is relatively underrepresented, the reconstruction of contextual state in the Amari-Hopfield network tends to infer contextual states based on the context domain rather than the stimulus domain. Consequently, it converges to an incorrect attractor that is not assigned to the current environmental state, thereby increasing perceptual error for external stimuli (hallucination-like effects). Moreover, SPE-driven remapping and the corresponding synaptic plasticity occur more frequently. In contrast, when the stimulus domain is overrepresented, the Amari-Hopfield network rarely assigns multiple contextual states to a given environmental state, leading to an overuse of default contextual states (see Figure 5B and Materials and methods).

Consistent with this prediction, when the stimulus domain is relatively underrepresented, agents fail to rapidly switch to the second experience of the environment I and II (Figure 5C). This failure is accompanied by an increased probability of context selections that differ from the true environmental state (hallucination-like effects). Moreover, the hallucination-like effects increase SPE-driven remapping, which occasionally leads to overlaps in context allocation in H (see Materials and methods), thereby accelerating the frequency of hallucination-like effects and leading to a decline in task performance. In contrast, when the stimulus domain is relatively overrepresented, persistent behavior is observed, and the correct rate in environment II becomes lower than environment I (Figure 5C). This is accompanied by an increased probability of default context usage due to failures in contextual state reconstruction (see Materials and methods) in environment II. Thus, our model predicts a relationship between sensory processing and behavioral flexibility in some psychosis.

Discussion

In this study, we proposed a simple, model-based reinforcement learning model equipped with two functional modules: Context selector and Sequence composer. We introduced two kinds of prediction error-based remapping, SPE-driven remapping and RPE-facilitated remapping as a key for generating context-dependent sequential activity change in hippocampus that enables flexible behavior. This mechanism is biologically plausible, as it is observed in the hippocampus (Bostock et al., 1991) and in some cortical regions (Castegnetti et al., 2021). Our model could simulate a variety of context-dependent sequential representations in hippocampus such as splitter cells (Wood et al., 2000), lap cells (Sun et al., 2020), probabilistic model selection (Ekman et al., 2022), and contextual inference (Julian and Doeller, 2021), without task-dependent parameter tuning. Furthermore, our model predicted a mechanistic explanation for the co-occurrence of deficits in sensory processing and flexible behavior. This result is supported by the clinical reports that psychosis can change the attractor dynamics in the hippocampus (Rolls, 2021) and treatments for sensory processing helped restore flexible behavior in some psychoses (Andelin et al., 2021; Javitt and Freedman, 2015; Pfeiffer et al., 2011; Reed et al., 2020). To the best of our knowledge, this is the first model that uses associative memory for describing the formation and switching of context-dependent hippocampal activity through remapping and its contribution to flexible behavior.

Our model is a functionally modular account of the cortical regions and hippocampus, enabling it to capture experimental findings across species. While hippocampal activity in rodents has been extensively characterized in terms of spatial coding, human hippocampal representations are more often non-spatial and episodic-like (Bellmund et al., 2018; Eichenbaum, 2017). For episodic memory to support flexible behavior, it would be beneficial to retrieve each episode in a context-dependent manner. The episodic contents may vary across species and individuals, yet the fundamental computations—estimating the current context from external stimuli and their

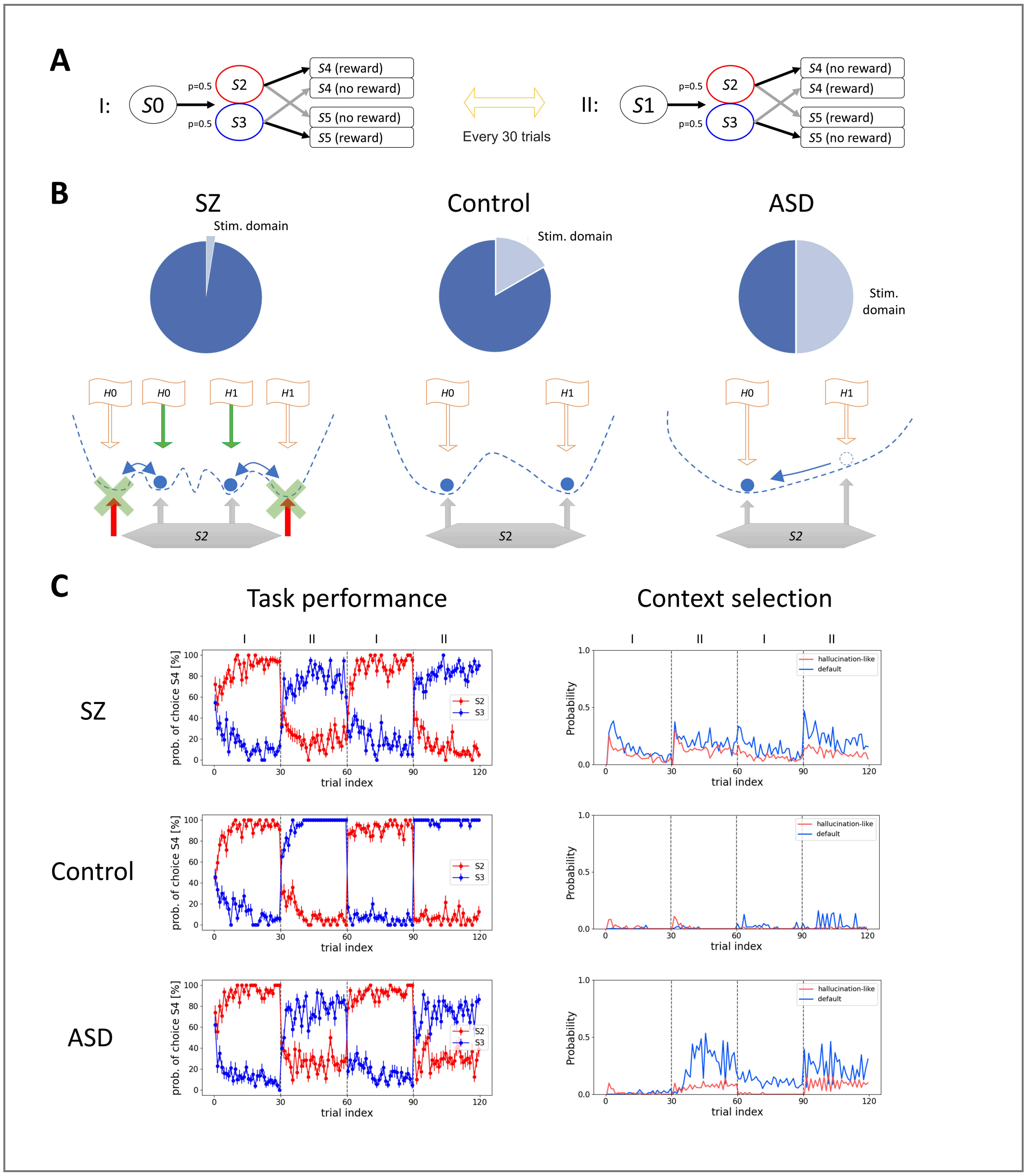

Figure 5. Model prediction about the relationship between sensory processing and flexible behavior.

A, Task diagram. The structure is the same as Figure 4, but the probability of S2 and S3 is equal. **B**, (Top) We tested three stimulus neuron ratios: 2.5% for SZ, 16.7% for control and 50% for ASD. (Bottom) Schematics of how Context selector changes by the manipulation of neuron ratios in this task. Blue dotted lines indicate the energy landscape and blue circles indicate the attractor dynamics. Red arrows indicate the wrong stimulus prediction (hallucination-like effects) which triggers SPE-driven remapping (green cross marks and arrows), and orange lines indicate the input from the hippocampus to X (*H*0 and *H*1 indicate hippocampal segments in S0 and S1, respectively). **C**, (Left) The probability of choosing S4 at S2 and S3 is plotted in red and blue, respectively. SZ model fails to show one-shot switch for the second experience of the environment I and II, while ASD model shows an impaired task performance mainly to the environment II. (Right) The result of context selection (see Figure S1). The probability of wrong stimulus reconstruction (hallucination-like) is plotted in red, and the probability of default context usage due to failures in context reconstruction (see Materials and methods) is plotted in blue.

history and flexibly updating this estimate via prediction errors—are likely conserved. Holding context information until the contextual prediction error is detected is analogous to the belief state in model-based reinforcement learning, which is known to improve performance under partially observable conditions (POMDPs) (Kaelbling et al., 1998). Our model provides a simple algorithmic implementation of this principle.

Although remapping is a widely known phenomenon, its mechanism remains under debate. We used the Amari-Hopfield network as Context selector to distinguish multiple contextual states that share the same external stimuli, and to reconstruct them via attractor dynamics from partial observations. We propose two advantages of this associative memory model. First, it can represent different contexts under the same external stimuli depending on the feedback from H to implement rapid behavioral switching without requiring synaptic changes. The second advantage is its ability to infer a contextual state using the associative memory mechanism. This property might occasionally yield a non-trivial contextual state based on past experiences. Expanding upon our model with more sophisticated associative memory search mechanisms could enable creative behavior.

We speculate that Context selector is implemented across multiple brain regions with varying degrees of resolution, including a part of the entorhinal cortex and prefrontal cortex. First, lateral EC (LEC) provides item-specific and sensory context information (Deshmukh and Knierim, 2011; Hargreaves et al., 2005), whereas the medial EC (MEC) supplies history information and state signals (Hafting et al., 2005; Heys and Dombeck, 2018). Because these inputs jointly shape hippocampal attractor dynamics, the EC is well positioned to determine which subjective context is selected. Second, PFC has been reported to retain context-dependent attractors, which reflect working memory (D'Ardenne et al., 2012), attention (Siegel et al., 2015), and confidence (Wynn and Nyhus, 2022), and to send inputs to the hippocampus. In addition, the PFC computes prediction errors that might trigger remapping.

Specifically, reward-related prediction errors are computed in the orbitofrontal cortex (OFC) (Garvert et al., 2023; Stalnaker et al., 2014), anterior cingulate cortex (ACC) (Seo and Lee, 2007) and ventromedial PFC (Rehbein et al., 2023), whereas stimulus-related prediction errors are calculated in the ACC (Ide et al., 2013) and dorsolateral PFC (Masina et al., 2018; Zmigrod et al., 2014). These neural circuits likely coordinate to estimate the current context and select the appropriate representation in the hippocampus via remapping. Our modeling of Context selector captures this core functionality in a simplified manner. Incorporating more elaborate features, such as multiple hierarchies (Rao, 2024), in future studies might help explain a broader range of experimental results.

Our model posits that the Sequence composer corresponds to computations within the hippocampus. As a biologically plausible projection, we consider the CA3–CA1 circuit, where contextual inputs from regions such as the PFC and EC provide the current contextual state to CA3, enabling the recurrent CA3–CA1 architecture to generate predictions of the next contextual state without errors in action. Consistent with this idea, the temporal lag in CA3 $\rightarrow$ CA1 transmission suggests a functional gradient in which CA3 represents present-oriented information while CA1 carries more future-oriented predictions (Chen et al., 2024), and neurons in both CA3 and CA1 exhibit action-driven remapping and encode action-planning signals (Green et al., 2022). Our framework, therefore, predicts that changes in CA3 $\rightarrow$ CA1 population activity precede behavioral switching in context-dependent alternation in Figure 2 or multi-lap tasks in Figure 3, and perturbation of this input will degrade the behavioral performance.

Beyond the function of individual components described above, our framework also yields several predictions about how these regions interact to support flexible behavior. We propose three experiments. First, our model posits that an error about the context triggers remapping. The OFC is known to be active when reward-related prediction error occurs (Banerjee et al., 2020), and hippocampal remapping is suggested to be induced by the entorhinal cortex, especially its lateral part (Latuske et al., 2017). Because a direct projection exists from the OFC to the lateral entorhinal cortex (Kondo and Witter, 2014), this input might critically influence hippocampal remapping. Second, our model suggests that the prediction error about the environment would induce a shift from place-cell encoding to lap-cell encoding in the hippocampus (Figure 3). Third, our model proposes two types of prediction error; one is the conventional prediction error that updates the synaptic weights within the context, and the other is the prediction error about the context that triggers remapping in X and H. How these two different prediction errors are represented in neural circuits will deepen our understanding of the neural basis of flexible behavior.

Our model also provides an algorithmic-level account of psychiatric symptoms by changing the relative weighting of sensory-encoding versus context-coding neurons. This implementation is analogous to Bayesian theories linking priors to psychiatric symptoms. In SZ, hallucinations and delusions have been modeled as arising from overly strong top-down priors (Powers et al., 2016) or circular inference, which leads to erroneous belief formation (Jardri et al., 2017; Jardri and Denéve, 2013). In our model, we used an underrepresented stimulus domain to increase the relative influence of internally generated context representation in context selection. Crucially, this implementation does not simply strengthen priors but induces excessive generation and competition of contextual states, leading to frequent yet non-reproducible remapping of hippocampal contextual activity and a failure of learning to converge despite repeated experience. In ASD, it has been argued that abnormally high sensory precision reduces the updating of expectations (Karvelis et al., 2018) or leads to sensory-dominant perception, which has been interpreted as weak priors (Angeletos, Chrysaitis and Series, 2023; Lawson et al., 2014; Pellicano and Burr, 2012). In our framework, we used an overrepresented stimulus domain to increase the relative influence of external stimulus representations in context selection. Importantly, our model captures not only sensory-dominant processing emphasized in previous studies, but also a distinctive impairment in flexibly utilizing newly introduced contexts, reflecting a failure of context reconstruction and resulting in persistent inflexible behavior. Thus, our conjunctive modeling of sensory and context processing complements Bayesian accounts of psychiatric symptoms and provides a mechanistic explanation for the role of sensory processing in maladaptive, inflexible behavior.

Our model also has limitations. First, there are context-dependent tasks that our model cannot solve. Although our model learns to separate contextual states, it does not combine them; consequently, we did not consider simulating the environment in which the number of hidden states decreases over time. Greater flexibility might be achieved by integrating both sensory and contextual information within certain neurons (e.g., Figure S3). Second, the resolution at which our model should distinguish different contextual states, including the stimulus resolution and time resolution, is hand-tuned in this work. While we used an abstract, grid-like state space with discrete time, an important direction for future work is to model its activity at finer-grained neural timescales, such as theta cycles (Foster and Wilson, 2007; Wikenheiser and Redish, 2015). In realistic, continuously changing environments, such resolutions should be adjusted autonomously. Introducing continuous and hierarchical representations with multiple levels of spatial and temporal resolution would facilitate such adjustments, potentially through mechanisms such as modern Hopfield networks (Krotov and Hopfield, 2020) or synfire-chain-based hippocampal sequence generation (Abeles, 1982; Diesmann et al., 1999; Shimizu and Toyoizumi, 2025; Toyoizumi, 2012), but this is beyond the focus of the current study. Third, our model assumed that only the hippocampus projects to the midbrain for reward prediction of sequential plans. However, there are projections from other brain regions, including the cortex, to the midbrain that are also involved in reward prediction (Jo and Mizumori, 2016). How these additional projections influence model-based behavior, especially in the case of hippocampal lesions, remains beyond the scope of this work. Finally, explicitly modeling the input from grid cells that encode geometric task structure (Krupic et al., 2015) might enable more sophisticated planning (e.g., discovering the shortest path).

Materials and methods

Simulation environment

We conducted all simulations and post-hoc analysis using a custom-made Python code. The source code is provided in Supplementary data.

Model description

Overview

Below, we introduce a model that describes the acquisition of model-based reasoning. Our model consists of two components: Context selector (X) and Sequence composer (hippocampus, H). For simplicity, the environment is defined in discrete time, and agents move through environmental states characterized by distinct external stimuli. The model operation relies on the environmental (behavioral) time step. At each time step, the agents perform contextual state estimation by Context selector and activate a corresponding hippocampal neuron. Then, this hippocampal neuron initiates sequential activity based on hippocampal synaptic connectivity. Each hippocampal sequence represents a planned course of action and is used to predict a series of external stimuli. The agents follow the plan unless SPE-driven remapping (see SPE-driven remapping section) or RPE-facilitated remapping (see RPE-facilitated remapping section). The hippocampal sequence from which actions are generated is updated upon a reward. After the action execution, the agents repeat the process by selecting the current contextual state. As the agents become familiar with the environment, hippocampal sequences that enable future predictions to become longer, and contextual state estimation by Context selector becomes less frequent. The algorithmic flow chart of our model is described in Figure S1.

Context selector (X)

We model Context selector as Amari-Hopfield network (Amari, 1972; Hopfield, 1982) of N = 1200 binary neurons, whose activity is described by vector X. We employ the Amari–Hopfield model because it allows multiple contexts to be stably maintained in response to stimuli and can be trained via Hebbian plasticity. We assume that similar computations are carried out in prefrontal and entorhinal cortical circuits in the brain.

X consists of two domains: stimulus domain X^{stim} and context domain X^{cont}. The neuron ratio in the stimulus domain over the whole neurons $\dim(X^{\text{stim}})/N$ is 16.7% for the control condition, 2.5% for the SZ condition, and 50% for the ASD condition. Note that dim describes the dimensions of a vector.

When the agents visit each environmental state for the first time, the X's activity is set to

$$X = \begin{pmatrix} X^{\text{stim}} \\ X^{\text{Cont}} \end{pmatrix} = \begin{pmatrix} \xi^{\text{stim}} \\ f\left(\xi^{\text{stim}}\right) \end{pmatrix} \quad \textbf{(eq.1)}$$

where converter function $f(\xi^{\text{stim}})$ = binary($A\xi^{\text{stim}} > a$) returns a binary vector computed from dim (X^{cont}) by dim (X^{stim}) default matrix A with independently and identically distributed unit Gaussian entries and scalar threshold a chosen so that $f(\xi^{\text{stim}})$ consists of half 1 and half 0 elements. This contextual state is set as a default context, ensuring that the X module assigns a unique contextual state to each environmental state. Biologically, one possible interpretation is that this default context corresponds to modality-specific innate representations in prefrontal regions (Manita et al., 2015).

From the second visit of each environmental state after completing actions according to a hippocampal sequence, the contextual state is determined by associative memory dynamics of the Amari-Hopfield network. We adopt two ways of initialization: history-based and landmark-based (see Figure S1). While the history-based initialization was introduced to select contextual state based on the history input from H, the landmark-based initialization was introduced to terminate the episodic sequence that would otherwise continue indefinitely. Biologically, the landmark-based initialization corresponds to the operation of anchoring a contextual state to salient environmental landmarks—such as an animal's nest—that serve as clear reference points. Formally, we use the history-based initialization when the input from H to X predicts the next contextual state, i. e. $W^{XH}H$ is not all zero, where W^{XH} represents the synaptic weights from H to X. We use the landmark-based initialization when the input from H to X does not provide any predictive input, i.e. W^{XH} is all zero, but they are at a landmark (we defined it as the initial environmental state of each task). When the inputs from H to X do not predict the next contextual state and agents are not at landmark (history mismatch), which typically happens after remapping, a new contextual state is generated and stored in the Amari-Hopfield network (see SPE-driven remapping and RPE-facilitated remapping section). In biologically, these distinctions could naturally arise from the interplay of the strength of history-dependent inputs, sensory saliency, and the depth of contextual attractors, which would be dynamically integrated in prefrontal and entorhinal cortical circuits.

The history-based initialization starts from the initial state of the Amari-Hopfield network

$$X_{\text{init}} = \text{binary}\left(W^{XH}H > 0\right) \tag{eq.2}$$

where binary represents the indicator function that takes 1 if the argument is true and 0 otherwise.

The landmark-based initialization starts from the initial state of the Amari-Hopfield network

$$X_{\text{init}} = \begin{pmatrix} \xi^{\text{stim}} \\ \text{random} \end{pmatrix} \tag{eq.3}$$

where random indicates a random binary vector consisting of half 0 and half 1 elements.

After history-based or landmark-based initialization, X is updated according to the associative memory dynamics:

$$X \leftarrow \text{binary}\left(W^{XX}\left(X - X^{0}\right) - \theta\right) \tag{eq.4}$$

where $\theta = 0.5$, $X^0 = 0.5$, and $\dim(X)$ by $\dim(X)$ matrix W^{xx} represents synaptic weights of Context selector (see Synaptic weight update section for how W^{xx} changes). These dynamics end up either as a successful or failed recall. A recall is defined as successful if X converges within 50 iterations, and its stimulus domain X^{stim} becomes identical to (ξ^{stim}. If X fails to converge within 50 iterations, the contextual state is set to the default contextual state defined in (*eq.* 1). This default implementation is analogous to psychological inertia, particularly under uncertainty (Ip and Nei, 2025; Sautua, 2017), which has been reported to be more pronounced in ASD patients (Joyce et al., 2017). If X converges within 50 iterations but the stimulus domain X^{stim} of the converged X is different from (ξ^{stim} (hallucination-like effects), agents consider that they are in a new context, and SPE-driven remapping occurs (see Figure 1S). Reuse of the default contextual state and the hallucination-like effects become critical for explaining ASD and SZ phenotypes, respectively. As one possible biological implementation, we consider that Context selection in X as the brain-wide evoked potential during which bottom-up information may be integrated with top-down signals to select the current context (Mohanty et al., 2025). In this case, it takes several hundred milliseconds for the contextual states in X to settle (Massimini et al., 2005).

After X is set, the agents randomly generate a hippocampal sequence reflecting it (see Sequence composer section). Then, the agents evaluate this sequence that encodes a course of actions and act according to it (see action flow section).

Sequence composer (hippocampus, H)

We model Sequence composer (hippocampus) with N = 300 binary recurrent neural network. The hippocampus produces sequential activity probabilistically based on the contextual state computed above. Starting from the seed hippocampal neuron directly activated by the contextual state, the next hippocampal neuron is iteratively activated with a probability proportional to the synaptic weights from the previously activated hippocampal neuron. Therefore, the same contextual state could generate diverse sequences. This randomness in the sequence generation facilitates the exploration behavior of the agents, which is important for reinforcement learning, but also adds noise to the input from Sequence composer to Context selector in the history-based computation.

Hippocampal neurons initially receive input vector $W^{HX}X_k$, where W^{HX} is the synaptic weight matrix from X to H, and X_k is the contextual state at time step k. Only the neuron that receives the strongest input is activated, whose index is described as

$$\widetilde{H}_k^{(S)} = \arg\max \left(W^{HX} X_k \right) \quad \text{(eq.5)}$$

(see Synaptic weight update section for how W^{HX} changes), where the tilde mark indicates a neuron index.

Our model has two types of hippocampal neurons: state-coding and transition-coding types. The indices of neurons belonging to these types are denoted as $\widetilde{H}^{(S)}$ and $\widetilde{H}^{(T)}$, respectively. The statecoding neurons receive input from X and represent the current contextual state, while the transitioncoding neurons send output to X and predict the next contextual state after an action i.e. T($X_{k+1}|X_k, a_{k,k+1}$). One possible biological grounding for this functional separation is that entorhinal cortex provide contextual inputs to CA3, and CA3 and CA1 generates predictions of next state through its recurrent architecture (Chen et al., 2024). Also, neurons in CA3 and CA1 are reported to show action-driven remapping to be involved in action planning (Green et al., 2022). When the agents experience a contextual state X_k for the first time, $\widetilde{H}_k^{(T)}$ is randomly chosen and the synaptic weight from $\widetilde{H}_k^{(S)}$ to $\widetilde{H}_k^{(T)}$ is set to 1. From the second experience of the contextual state X_k, the corresponding hippocampal neuron $\widetilde{H}_k^{(S)}$ initiates a sequence $H = \left[\widetilde{H}_k^{(S)}, \widetilde{H}_k^{(T)}, \ldots, \widetilde{H}_{k+\tau}^{(S)}, \widetilde{H}_{k+\tau}^{(T)}\right]$ of hippocampal activity with a non-negative integer r, where the next neuron is recursively chosen with a probability vector proportional to

$$\frac{[W^{HH}]_{\cdot \widetilde{H}_k} - w_0}{1 - w_0} \text{binary} \left([W^{HH}]_{\cdot \widetilde{H}_k} - w_0 > 0.01 \right) \quad \text{(eq.6)}$$

where $[W^{HH}]_{\cdot \widetilde{H}_k}$ describes a vector of intra-hippocampal synaptic weights from neuron $\widetilde{H}_k$ and w_0 = 0.3 is the effective threshold. The sequential activity can stop at a transition-coding neuron $\widetilde{H}_{k+\tau}^{(T)}$ according to two conditions: when all the synaptic weights $[W^{HH}]_{\cdot \widetilde{H}_{k+\tau}^{(T)}}$ are equal to or below 0.01 or when the reward value function of the lastly activated transition-coding neuron $\widetilde{H}_{k+\tau}^{(T)} = \widetilde{H}_{-1}$ becomes positive (see Reward prediction section).

The synaptic connection from a state-coding neuron to a transition-coding neuron is formed in a reward-independent manner as described above, whereas the connection from a transition-coding neuron to a state-coding neuron is established in a reward-dependent manner (see Synaptic Weight Update section). Consequently, when animals receive few rewards during the initial exploration phase, minimal sequences with τ = 0 are constructed. As animals discover rewarding behaviors, these minimal sequences join, and eventually, agents anticipate the rewarding transition ahead.

When the number of contextual states increases particularly in the SZ condition, representational overlap arises between hippocampal state-coding and transition-coding neurons. This overlap makes the prediction of the next contextual state by the transition-coding neurons unreliable. The degraded prediction from H, in turn, corrupts the initial condition for context selection in X (Eq. 3), leading to hallucination-like behavior.

Reward prediction

Each hippocampal sequence $\mathscr{H}$ is associated with rewards, perhaps via the operation of the midbrain. Reward value function $V_{\widetilde{H}_{-1}}$, which depends on the lastly activated transition-coding hippocampal neuron $\widetilde{H}_{-1}$ of the sequence, is updated every time the agents receive reward $R > 0$ according to

$$V_{\widetilde{H}_{-1}} \leftarrow V_{\widetilde{H}_{-1}} + \alpha \left(R - V_{\widetilde{H}_{-1}} \right) \quad \textbf{(eq.7)}$$

with learning rate $\alpha = 0.15$. The sequence value $SV_{\widetilde{H}_{-1}}$ associated with $\widetilde{H}_{-1}$ mirrors $V_{\widetilde{H}_{-1}}$ except when it is suppressed by this neuron's *no-good* indicator $NG_{\widetilde{H}_{-1}}$ (cross marks in Figure 1C), namely,

$$SV_{\widetilde{H}_{-1}} = V_{\widetilde{H}_{-1}} - \left(V_{\widetilde{H}_{-1}} + NG_{\widetilde{H}_{-1}} \right) \cdot \text{binary} \left(NG_{\widetilde{H}_{-1}} \geq \theta_{NG} \right) \quad \textbf{(eq.8)}$$

where suppression threshold θ_{NG} is set to 0.7. *No-good* indicator is introduced to transiently suppress previously established sequences that have not been recently rewarded, without devaluing them. This *no-good* indicator facilitates RPE-facilitated remapping (see RPE-facilitated remapping section) that leads to exploration of different contextual states in X and sequences in H. The *no-good* indicator is inspired by recent findings in the ventral hippocampus, where dopamine D2-expressing neurons of the ventral subiculum selectively promote exploration under anxiogenic contexts (Godino et al., 2025). When the *no-good* indicator is active, i.e. $NG_{\widetilde{H}_{-1}} \geq \theta_{NG}$, the sequence value becomes transiently negative. Note that we set θ_{NG} as 0.7 to make the agents sufficiently sensitive to abrupt environmental changes and enable exploring some candidate contexts after RPE-facilitated remapping.

These neurons' *no-good* indicators change when a reward is presented. The *no-good* indicator of the lastly activated hippocampal neuron $\widetilde{H}_{-1}$ instantaneously drops to 0 when the reward is greater than the reward value function, i.e., $R > V_{\widetilde{H}_{-1}}$ but instantaneously increases by 1 otherwise. In addition, the *no-good* indicators of all hippocampal neurons gradually decay according to

$$NG \leftarrow \gamma NG \quad \textbf{(eq.9)}$$

with multiplication factor $\gamma = 0.7$ when the reward is less than the reward value function, i.e., $R < V_{\widetilde{H}_{-1}}$.

Action flow

After completing each environmental state transition according to a planning sequence without remapping (see SPE-driven remapping and RPE-facilitated remapping section), the agents estimate a contextual state (context selection, Figure S1) and, based on it, generate a hippocampal sequence $\mathscr{H}$ (sequence composition, Figure S1). Below, we describe how the agents select one hippocampal sequence. The last hippocampal neuron $\widetilde{H}_{-1}$ of the sequence $\mathscr{H}$ informs its sequence value $SV_{\widetilde{H}_{-1}}$ (see Reward prediction section). When $SV_{\widetilde{H}_{-1}}$ is positive, the agents select this sequence. Otherwise, the agents reject this hippocampal sequence and compose another hippocampal sequence (using a different random seed for the landmark-based initialization) for up to nine attempts (sequence selection loop, Figure S1). If none of the nine sequences have positive $SV_{\widetilde{H}_{-1}}$, one is selected randomly, excluding that with the lowest sequence value. We use this sequence selection to provide a balance between exploration and exploitation of sequence selection and serve as a good compromise for visualization. Once a sequence is selected, agents start to execute sequence-based action loop (see Figure S1) unless RPE-facilitated remapping (see RPE-facilitated remapping section). The transition-coding hippocampal neurons $\left[\widetilde{H}_k^{(T)}, \ldots, \widetilde{H}_{k+\tau}^{(T)} \right]$ in the sequence specify the transition of environmental states (action), i.e. the stimulus domain of the input from H to X (binary($W^{XH}H^{(T)} > 0$)) represents the prediction of the next environmental states, where W^{XH} is the synaptic weights from H to X and $H^{(T)}$ is the hippocampal state when each transition-coding neuron is active. The sequence with τ transitioncoding neurons provides an action plan in the next τ steps, unless SPE-driven remapping happens (see SPE-driven remapping section). This is inspired by preplay or planning by hippocampal sequences (Dragoi and Tonegawa, 2011). After completing the final action specified by the sequence, the agents repeat the whole procedure, starting from the contextual state estimation.

SPE-driven remapping

SPE-driven remapping can occur while the agents execute a course of actions following a hippocampal sequence. We refer to SPE-driven remapping as the shift of *X*'s activity to another contextual state or generate a new one under the same external stimuli. Upon the course of actions following hippocampal sequence $\mathscr{H} = \left[\widetilde{H}_k^{(S)}, \widetilde{H}_k^{(T)}, \ldots, \widetilde{H}_{k+\tau}^{(S)}, \widetilde{H}_{k+\tau}^{(T)}\right]$, the prediction of the next external stimuli, i.e. the stimulus domain of binary($W^{XH}H^{(T)} > 0$), may differ from the actual one, ξ^{stim}. When this happens at a sequence location $k + \tau'$ ($1 \le \tau' \le \tau$), SPE-driven remapping (Figure 1D) occurs, and the synaptic weights in X and H are modified. If the event occurs during $1 \le \tau' < \tau$, steps 1–3 are applied. In contrast, if the event occurs at $\tau' = \tau$, only step 3 is applied.

1. The hippocampal sequence is interrupted between the transition $\widetilde{H}_{k+\tau'-1}^{(T)} \to \widetilde{H}_{k+\tau'}^{(S)}$, and the corresponding synaptic weight is weakened (see Synaptic weight update section).
2. If the transition-coding neuron $\widetilde{H}_{k+\tau'-1}^{(T)}$ projects to state-coding neurons other than $\widetilde{H}_{k+\tau'}^{(S)}$, these state-coding neurons' predictions about external stimuli are examined. If there exists one that predicts the actual external stimuli with an error less than the remapping threshold of $\theta_{remap} = 5$ *bit*, this neuron is activated, and the contextual state *X* is set based on its input (eq. 2, Context selector section). Otherwise, step 3 is applied. Note that we set the remapping threshold $\theta_{remap} = 5$ *bit* to allow for small miss-convergence during recall in the Amari–Hopfield model.
3. A new contextual state is set as $X = (\xi^{\text{stim}}, \text{random})$ with the synaptic weights W^{xx} updated (see Synaptic weight update section).A hippocampal neuron is activated based on the new contextual state in *X* following *eq.* 5, and the synaptic weight is strengthened between the interrupted transition-coding hippocampal neuron $\widetilde{H}_k^{(T)}$ and the newly activated state-coding hippocampal neuron (see Synaptic weight update section).

When a new hippocampal neuron is recruited in step 3, the history mismatch occurs in the following environmental state because this hippocampal neuron does not predict upcoming external stimulus. Therefore, once SPE-driven remapping is triggered, the contextual states in X as well as the activated neurons in H are repeatedly updated in the following environmental states until the agents encounter a landmark (i.e. starting point) and reset the episode.

RPE-facilitated remapping

To gain information on the environment, the agents perform exploration followed by RPE-facilitated remapping. We refer to exploration as a random action not specified by the selected sequence. Exploration can occur with probability p_{expl} whenever the agents enter an environmental state with the number of transition candidates greater than the number of transition-coding hippocampal neurons initiating from the corresponding state-coding hippocampal neuron. The exploration probability is generally $p_{expl} = 0.3$ but increases to certainty ($p_{expl} = 1$) if the agents are taking actions following a sequence with a negative sequence value, which happens when its *no-good* indicator is active, i.e. $NG_{\widetilde{H}_{-1}} \ge \theta_{NG}$. In case of this exploration, one of the unconnected transition-coding hippocampal neurons is randomly activated (RPE-facilitated remapping), and the agents take a random transition. At the following environmental state, X is set to be a random contextual state $X = (\xi^{stim}, random)^T$, and synaptic weights of H and X are updated (see Synaptic weight update section).

Same as SPE-driven remapping, once RPE-facilitated remapping is triggered, the history mismatch occurs and the contextual states in X as well as the activated neurons in H are repeatedly updated in the following environmental states until the agents encounter a landmark (i.e. starting point) and reset the episode.

Synaptic weight update

We used a Hebbian learning rule to update the synaptic weight matrix W^{XX} only for the first time contextual state X is settled:

$$W^{XX} \leftarrow W^{XX} + \left(X - X^{0}\right)\left(X - X^{0}\right)^{\mathrm{T}} \tag{eq.10}$$

We also used a basic Hebbian learning rule for updating synaptic weights between X and H. Again, only for the first time a hippocampal neuron is activated according to (eq. 5) in response to contextual state X_k, synaptic weights are updated as

$$W^{HX} \leftarrow W^{HX} + \eta H^{(S)}\left(X_k - X^{1}\right)^{\mathrm{T}} \tag{eq.11}$$

$$W^{XH} \leftarrow W^{XH} + \eta\left(X_k - X^{1}\right)\left(H^{(S)}\right)^{\mathrm{T}} \tag{eq.12}$$

$$W^{XH} \leftarrow W^{XH} + \eta\left(X_k - X^{1}\right)\left(H_{-1}^{(T)}\right)^{\mathrm{T}} \tag{eq.13}$$

where $H^{(s)}$ and $H^{(T)}$ are the state-coding and transition-coding hippocampal activity vectors, respectively, whose elements take 1 for the activated neuron of the corresponding type and 0 for the others. Note that the initial synaptic weights of W^{HX} and W^{XH} are all 0. Similarly, $H_{-1}^{(T)}$ is the transition-coding hippocampal activity vector of the previous hippocampal sequence, where the element corresponding to the last transition-coding neuron takes 1, and others take 0. Learning rate η = (N + 1)/2 and offset $X^1 = N/(N+1)$ are chosen to achieve good association dynamics in Context selector. These synaptic weights change within the bound W^{XH}, $W^{HX} \leq 1/2$.

We used different learning rules for the intra-hippocampal synaptic weights depending on within-episodic and between-episodic segments. The initial synaptic weights are all w_0, and these weights change within the bound $0 \leq W^{HH} \leq 1$. Within-episodic connections, i.e. state-coding to transitioncoding synapses, are constantly updated in a reward-independent manner when and $\widetilde{H}_k^{(S)}$ are $\widetilde{H}_k^{(T)}$ activated as

$$\begin{aligned} W^{HH} \leftarrow\ & W^{HH} + H_k^{(T)}\left(H_k^{(S)}\right)^{T} - w_0 \\ & - 0.5\alpha H_k^{(T)}\left(1 - H_k^{(S)}\right)^{T} \mathrm{binary}\left(\left[W^{HH}\right]_{\widetilde{H}_k^{(T)}\widetilde{H}_k^{(S)}} \leq w_0\right) \end{aligned} \tag{eq.14}$$

The second term describes Hebbian potentiation, and the third term describes hetero-synaptic depression between non-active presynaptic neurons and the active postsynaptic neuron. Note that we assume hetero-synaptic depression only upon the initial establishment of the synaptic connection between the two hippocampal neurons. This modeling is inspired by behavioral time scale plasticity in the hippocampus (Bittner et al., 2017), in which synaptic potentiation occurs for events that are close in time regardless of reward, and such plasticity is believed to support the formation of place cells etc.. Between-episodic connections, i.e. transition-coding to state-coding synapses, are constantly updated in a reward-dependent manner when the agents receive a reward (R > 0) and $\widetilde{H}_k^{(T)}$ and $\widetilde{H}_{k+1}^{(S)}$ are involved in $\mathscr{H}$ according to

$$W^{HH} \leftarrow W^{HH} + \alpha\left(R - [W^{HH}]_{\widetilde{H}^{(S)}_{k+1}\widetilde{H}^{(T)}_{k}} - w_0\right) H^{(S)}_{k+1}\left(H^{(T)}_{k}\right)^{\mathrm{T}}$$
$$-0.5\alpha H^{(S)}_{k+1}\left(1 - H^{(T)}_{k}\right)^{\mathrm{T}} \mathrm{binary}\left([W^{HH}]_{\widetilde{H}^{(S)}_{k+1}\widetilde{H}^{(T)}_{k}} \leq w_0\right) \quad \text{(eq.15)}$$

The second term describes Hebbian potentiation that modifies the weight $[W^{HH}]_{\widetilde{H}^{(S)}_{k+1}\widetilde{H}^{(T)}_{k+\tau}}$ toward R — w_0, and the third term describes hetero-synaptic depression. This is supported by the finding that dopaminergic neuromodulation gates LTP, enabling preferential consolidation of reward-associated experiences (Lisman and Grace, 2005; Takeuchi et al., 2016).

In addition, if SPE-driven remapping happens at the sequence location between $\widetilde{H}^{(T)}_{k}$ and $\widetilde{H}^{(S)}_{k+1}$, the synaptic weight from $\widetilde{H}^{(T)}_{k}$ to $\widetilde{H}^{(S)}_{k+1}$ is weakened by — $\alpha[W^{HH}]_{\widetilde{H}^{(S)}_{k+1}\widetilde{H}^{(T)}_{k+\tau}}$, while that from $\widetilde{H}^{(T)}_{k}$ to the activated state-coding hippocampal neurons $\widetilde{H}^{(S)}_{k+1}$, is strengthened by α (0.65 – $[W^{HH}]_{\widetilde{H}^{(S)}_{k+1},\widetilde{H}^{(T)}_{k}}$) binary ($[W^{HH}]_{\widetilde{H}^{(S)}_{k+1},\widetilde{H}^{(T)}_{k}}$ < 0.65).

Considering the memory capacity of the Amari-Hopfield Network with correlated patterns, the number of memorizable contextual states sharing the same external stimulus is below 8. If this condition is violated, to prevent overloading the Amari-Hopfield network, the contextual state X that has never produced hippocampal sequences with a sequence value more than 0.7 induces a forgetting process as

$$W^{XX} \leftarrow W^{XX} - \left(X - X^0\right)\left(X - X^0\right)^{\mathrm{T}} \quad \text{(eq.16)}$$

This process represents forgetting of reward-unrelated episodic memory.

Formal descriptions of each task setting

All tasks used in this study were formulated as partially observable Markov decision processes (POMDPs), defined as

$$M = \langle S, A, T, R, C\rangle$$

Below, we describe the model components for each task.

Alternation task (Figure 2)

The alternation task (Figure 2) can be described as follows.

- **State space** $S = \{S_1, S_2, S_3, S_4, S_5\}$, where S_1 is the starting point, and S_4 and S_5 are the reward delivery points.
- **Action space** $A = \{a_{12}, a_{24}, a_{25}, aa_{32}, a_{43}, a_{51}\}$, where each action determines a state transition.
- **Transition function** $T(S_j|S_j, a_{ij}) = 1$, specifying the probability of reaching state S_j given current state S_i and action a_{ij}. In this task, transitions are deterministic given the correct context, but ambiguous without context.
- **Reward function** $R_t(s, c_t) = \begin{cases} 1 \text{ if } (s = S_4, c_t = 1) \\ 1 \text{ if } (s = S_5, c_t = 2) \\ \quad 0 \quad otherwise \end{cases}$, where t indicates the trial index, and c_t indicates the hidden state at trial t.
- **Hidden state** $c_t = \begin{cases} 1 \text{ if } c_{t-1} = 2 \text{ and } R_{t-1} = 1 \\ 2 \text{ if } c_{t-1} = 1 \text{ and } R_{t-1} = 1 \end{cases}$, where hidden variable switches depending on the previous reward under initial condition of $c_1 = 1$.

2-laps task (Figure 3)

2-laps task (Figure 3) can be described as follows.

- **State space** $S = \{S_1, S_2, S_3, S_4\}$, where S_1 is the starting point, and S_4 is the reward delivery point.

- **Action space** $A = \{a_{12}, a_{23}, a_{24}, a_{32}\}$, where each action determines a state transition.
- **Transition function** $T(S_j|S_i, a_{ij}) = 1$, specifying the probability of reaching state S_j given current state S_i and action a_{ij}. In this task, transitions are deterministic given the correct context, but ambiguou s with out context.
- **Reward function** $R_t(s, c_t) = \begin{cases} 1 \text{ if } \left(s = S_4, N_t(S_3) = 0, \quad c_t = 1\right) \\ 1 \text{ if } \left(s = S_4, N_t(S_3) = 1, \quad c_t = 2\right) \\ 1 \text{ if } \left(s = S_4, N_t(S_3) \geq 2, \quad c_t = 3\right) \\ \quad 0 \quad otherwise \end{cases}$, where t indicates the trial index, N_t ($S3$) indicates the number of visiting $S3$ at trial t, and c_t indicates the hidden state at trial t.
- **Hidden state** $c_t = \begin{cases} 1 \text{ if } t \leq 20 \\ 2 \text{ if } 20 < t < 40 \\ 3 \text{ if } t \geq 40 \end{cases}$, where hidden variable switches depending on the trial index.

Note that in Figure 3G-H we used the following Reward function and Hidden state.

- **Reward function** $R_t(s, c_t) = \begin{cases} 1 \text{ if } \left(s = S_4, N_t(S_3) = 1, \quad c_t = 2\right) \\ 1 \text{ if } \left(s = S_4, N_t(S_3) \geq 2, \quad c_t = 3\right) \\ \quad 0 \quad otherwise \end{cases}$, where t indicates the trial index, N_t ($S3$) indicates the number of visiting $S3$ at trial t, and c_t indicates the hidden state at trial t.
- **Hidden state** $c_t = \begin{cases} 2 \ if \ \mathrm{mod} \ (t, 60) \leq 30 \\ 3 \quad other\ wise \end{cases}$, where hidden variable switches depending on the trial index.

Simplified probabilistic cueing task (Figure 4 and 5)

Simplified probabilistic cueing task (Figure 4 and 5) can be described as follows.

- **State space** $S = \{S_0, S_1, S_2, S_3, S_4, S_5\}$, where S_0 (if $c_t = 1$)or S_1 (if $c_t = 2$) are the starting points and S_4 and S_5 are the reward delivery points.
- **Action space** $A = \{a_{0(23)}, a_{1(23)}, a_{24}, a_{25}, a_{34}, a_{35}\}$, where each action determines a state transition.
- **Transition function** $T(S_j|S_i, a_{ij}) = \begin{cases} 1 - p \text{ if } (i, f) \in \{(0, 3), \ (1, 2)\} \\ \quad p \text{ if } (i, f) \in \{(0, 2), \ (1, 3)\} \\ \quad 1 \quad otherwise \end{cases}$, specifying the probability of reaching state S_j given current state S_i and action a_{ij}. We set $p = 0.8$ in Figure 4, and we set $p = 0.5$ in Figure 5.
- **Reward function** $R_t(s, a, c_t) = \begin{cases} 1 \text{ if } \left(s = S_4, a = a_{24}, \ c_t = 1\right) \\ 1 \text{ if } \left(s = S_5, a = a_{35}, \ c_t = 1\right) \\ 1 \text{ if } \left(s = S_4, a = a_{34}, \ c_t = 2\right) \\ 1 \text{ if } \left(s = S_5, a = a_{25}, \ c_t = 2\right) \\ \quad 0 \ otherwise \end{cases}$, where t indicates the trial index, and c_t indicates the hidden state at trial t.
- **Hidden state** $c_t = \begin{cases} 1 \ if \ \mathrm{mod} \ (t, 40) \leq 20 \\ 2 \quad other\ wise \end{cases}$, where hidden variable switches depending on the trial in dex.

Model-free learning with temporal contexts

To highlight the advantage of our model, we compared it to the Q-learning with temporal contexts (Figure S2), namely, the state is defined by the recent n-step history of environmental state (i.e., $s_k^{(n)} = (S_k, S_{k-1}, ..., S_{k-n})^{\mathrm{T}}$, where $s_k^{(n)}$ is the temporal context state, and S_k is the environmental state at time k). We changed n from 0 to 3. In the Q-learning, the action value for a temporal state s_k to the next s_{k+1} is updated as

$$Q(s_k, s_{k+1}) \leftarrow (1 - \alpha) Q(s_k, s_{k+1}) + \alpha \left(R(s_{1:k+1}) + \gamma \max_s Q(s_{k+1,} s) \right) \quad \textbf{(eq.17)}$$

where the initial Q value is 0, learning rate $\alpha = 0.4$, the discount factor $\gamma = 0.6$ and the task dependent reward function $R = 100$ for the rewarded transition and $R = 1$ for else. Next state selection policy π is set to be proportional to Q value as

$$\pi\left(s_k, s_{k+1}\right) \propto Q\left(s_k, s_{k+1}\right) \tag{eq.18}$$

Inhibition experiment

To replicate the inhibition experiment of medial entorhinal cortex axons at CA1, we inhibit 98.5% of the input from the context domain of X to H. After the 2-laps task in Figure 3, we observed the hippocampal activity responding to each contextual state with or without this inhibition. ESR correlation is calculated based on the hippocampal activity of each lap, while the spatial correlation is calculated based on that of space. To avoid nan value when calculating correlations, we assumed that the activity of hippocampal cells without firing would have a random spontaneous activity between 0 and 0.1. Note that this operation does not significantly affect the result.

Supplementary figures

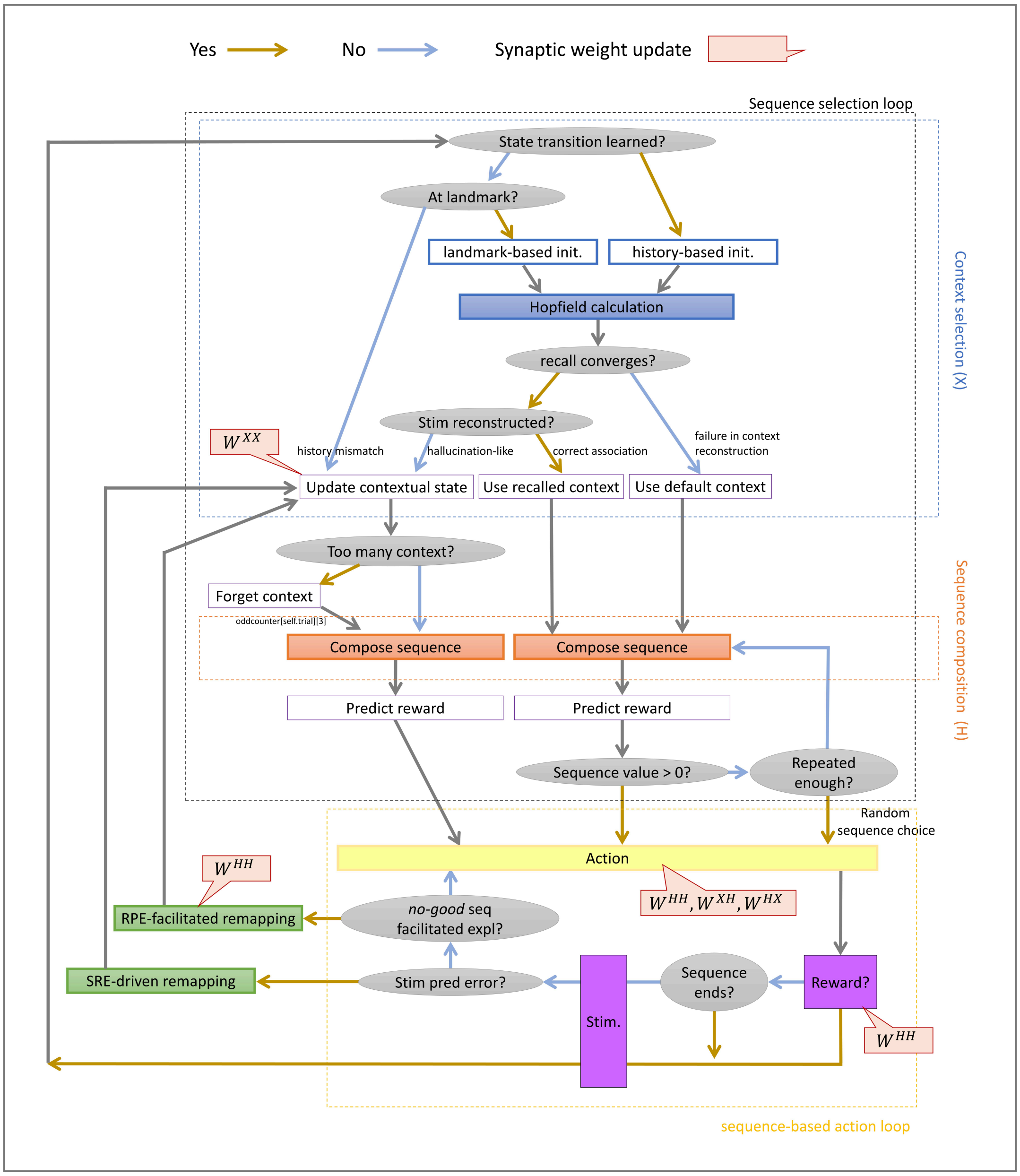

Figure S1. The algorithmic flow chart of the model. Square boxes show the manipulation explained in Materials and methods, while the gray circles show if bifurcation with yes for ochre arrows and no for blue arrows. Synaptic weight updates are indicated in the pink boxes. Context selection in X is indicated in the blue dotted box, and sequence composition in H is indicated in the orange dotted box. The black dotted box indicates the sequence selection through the interaction between X and H, and the yellow dotted box indicates the action loop after the sequence selection.

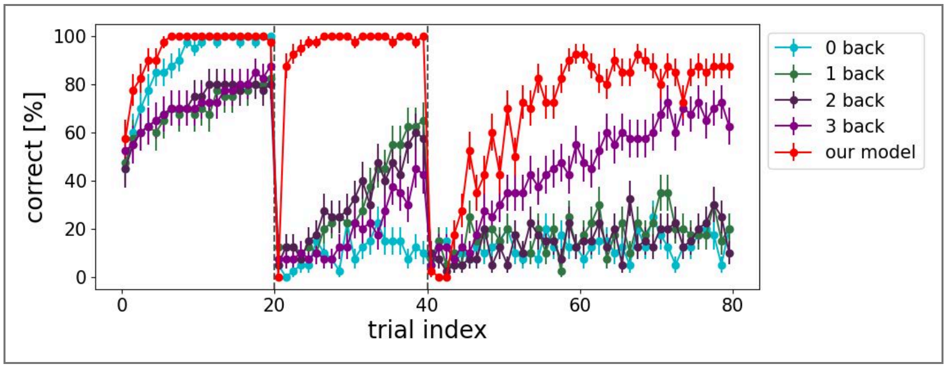

Figure S2. **2-laps task with model-free learning with temporal contextual states.**

The contextual states are defined by the composition of the current state and n back sensory histories. It requires at least 3 back histories to complete this task, but the correct rate of 3 back histories is worse than our model.

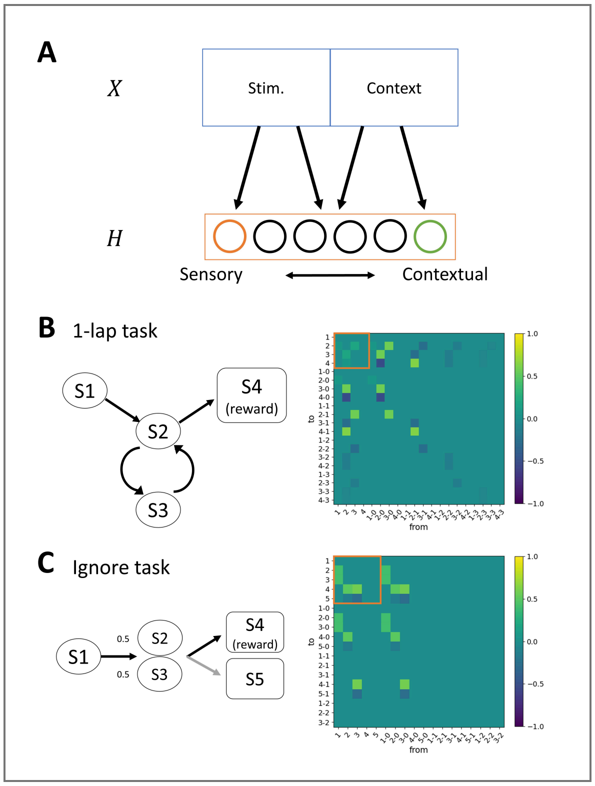

Figure S3. **Reward-dependent plasticity when sensory and contextual encoding neurons coexist in hippocampus.**

A, Schematic figure of how sensory and contextual encoding neurons can coexist in the hippocampus. Hippocampal neurons that receive synaptic input mainly from the stimulus-encoding region have sensory encoding, while those from the context-encoding region have contextual encoding. **B**, How the hippocampal network evolves when sensory and contextual encoding neurons coexist in the 1-lap task. This task requires contextual encoding, otherwise agents cannot distinguish between the first and second visit of S2. After 100 trials of random exploration in this area, the network between sensory encoding hippocampal neurons (indicated by the orange square) does not increase synaptic weights, while that between relevant context-encoding hippocampal neurons increases synaptic weights. **C**, How the hippocampal network evolves when sensory and contextual encoding neurons coexist in the ignore task. In this task, contextual encoding is not necessary because agents receive a reward at S4 independent of past states or latent variables. In contrast to the 1-lap task, the network between sensory encoding hippocampal neurons (indicated by the orange square) increases the synaptic weights as well as that between context encoding hippocampal neurons.

Data availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials. All source code is provided in *https://github.com/toppo365/flexiblemodel.git*.

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Peer reviews

Reviewer #2 (Public review):

Summary:

Ito and Toyoizumi present a computational model of context-dependent action selection. They propose a "hippocampus" network that learns sequences based on which the agent chooses actions. The hippocampus network receives both stimulus and context information from an attractor network that learns new contexts based on experience. The model is consistent with a variety of experiments both from the rodent and the human literature such as splitter cells, lap cells, the dependence of sequence expression on behavioral statistics. Moreover, the authors suggest that psychiatric disorders can be interpreted in terms of over/under representation of context information.

My general assessment of the work is unchanged, and I still have some questions requesting methodological clarification

Strengths:

This ambitious work links diverse physiological and behavioral findings into a self-organizing neural network framework. All functional aspects of the network arise from plastic synaptic connections: Sequences, contexts, action selection. The model also nicely links ideas from reinforcement learning to a neuronally interpretable mechanisms, e.g. learning a value function from hippocampal activity.

Weaknesses:

The presentation, particularly of the methodological aspects, needs to be heavily improved. Judgment of generality and plausibility of the results is severely hampered but is essential, particularly for the conclusions related to psychiatric disorders. In its present form, it is impossible to judge whether the claims and conclusions made are justified. Also, the lack of clarity strongly reduces the impact of the work on the field.

Comments:

The authors have made strong efforts to improve on their description of the methods, however, it is still very hard to understand. As a result of some of their clarifications, new issues appeared that I was not able to extract in the previous version.

(1) Particularly I had problems figuring out how the individual dynamical systems are interrelated (sequences, attractor, action, learning). As I understand it now (and I still might be wrong) there is one discrete time dynamics, where in each time step one action takes place as well as the attractor and sequence dynamics are moved one step forward. Also, synaptic updates happen in every one of those time steps. The authors may verify or correct my interpretations and further improve on their description in the manuscript. It is also confusing that time in the figure panels is given in units of trials, where each trial may consist of (maybe different amounts of) multiple time steps. Are the thin horizontal red ad blue lines time steps?

(2) As a consequence of my new understanding of the model dynamics, I have become doubts about the interpretation of the attractor network as context encoding. Since the X population mainly serves to disambiguate sequence continuation, right before the action has to be taken (active for only two time steps in Figure 1C?) they could also be considered to encode task space (El-Gaby et al. 2024; doi: 10.1038/s41586-024-08145-x).

(3) Also technically, I wonder why the authors introduce the criterion of 50(!) time steps to allow the attractor to converge, if the state of the attractor network is only relevant in one time step to choose the appropriate continuation of the sequence of actions. Is attractor dynamics important at all? What would happen if just the input and output weights to the X population are kept and the recurrent weights are set 0?

(4) Figure 3E: How many time steps are the H cells active (red bars?) Figure 4J: What are the units of the time axis?

https://doi.org/10.7554/eLife.106506.2.sa2

Reviewer #3 (Public review):

Summary:

This paper develops a model to account for flexible and context-dependent behaviors, such as where the same input must generate different responses or representations depending on context. The approach is anchored in the hippocampal place cell literature. The model consists of a module X, which represents context, and a module H (hippocampus), which generates "sequences". X is a binary attractor RNN, and H appears to be a discrete binary network, which is called recurrent but seems to operate primarily in a feedforward mode. H has two types of units (those that are directly activated by context, and transition/sequence units). An input from X drives a winner-take-all activation of a single unit H_context unit, which can trigger a sequence in the H_transition units. When a new/unpredicted context arises, a new stable context in X is generated, which in turn can trigger a new sequence in H. The authors use this model to account for some experimental findings, and on a more speculative note, propose to capture key aspects of contextual processing associated with schizophrenia and autism.

Strengths:

Context-dependency is an important problem. And for this reason, there are many papers that address context-dependency - some of this work is cited. To the best of my knowledge, the approach of using an attractor network to represent and detect changes in context is novel and potentially valuable.

Comments on revisions:

The authors have adequately addressed my concerns. Most importantly, the details of the implementation of the different components of the model are much more clearly described.

https://doi.org/10.7554/eLife.106506.2.sa1

Author response:

The following is the authors' response to the original reviews.

> ***eLife Assessment***
>
> *This is a potentially valuable modeling study on sequence generation in the hippocampus in a variety of behavioral contexts. While the scope of the model is ambitious, its presentation is incomplete and would benefit from substantially more methodological clarity and better biological justification. The work will interest the broad community of researchers studying corticalhippocampal interactions and sequences.*

Thank you very much for your comments. We are very encouraged by your positive feedback. We have revised our manuscript to clarify our model, strengthen its biological justification, and make it more accessible to a broader audience.

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *The manuscript by Ito and Toyozumi proposes a new model for biologically plausible learning of context-dependent sequence generation, which aims to overcome the predefined contextual time horizon of previous proposals. The model includes two interacting models: an Amari-Hopfield network that infers context based on sensory cues, with new contexts stored whenever sensory predictions (generated by a second hippocampal module) deviate substantially from actual sensory experience, which then leads to hippocampal remapping. The hippocampal predictions themselves are context-dependent and sequential, relying on two functionally distinct neural subpopulations. On top of this state representation, a simple Rescola-Wagner-type rule is used to generate predictions for expected reward and to guide actions. A collection of different Hebbian learning rules at different synaptic subsets of this circuit (some reward-modulated, some purely associative, with occasional additional homeostatic competitive heterosynaptic plasticity) enables this circuit to learn state representations in a set of simple tasks known to elicit context-dependent effects.*

We appreciate it for carefully reading the manuscript and finding the novelty and significance in our work.

> *Strengths:*
>
> *The idea of developing a circuit-level model of model-based reinforcement learning, even if only for simple scenarios, is definitely of interest to the community. The model is novel and aims to explain a range of context-dependent effects in the remapping of hippocampal activity.*
>
> *Weaknesses:*

> *The link to model-based RL is formally imprecise, and the circuit-level description of the process is too algorithmic (and sometimes discrepant with known properties of hippocampus responses), so the model ends up falling in between in a way that does not fully satisfy either the computational or the biological promise. Some of the problems stem from the lack of detail and biological justification in the writing, but the loose link to biology is likely not fully addressable within the scope of the current results. The attempt at linking poor functioning of the context circuit to disease is particularly tenuous.*

We thank the reviewer for the insightful comments.

To better characterize our model, we added formal descriptions of each task setting and explicitly specified the sources of uncertainty. We revised the schematic figures in Figure 1 to more clearly illustrate our model. An important revision is that we now distinguish between stimulus prediction error (SPE)–driven remapping and reward prediction error (RPE)–facilitated remapping. SPEdriven remapping is triggered by mismatches between actual sensory stimuli and those predicted from past history and serves to update the current contextual state or to create a new one. In contrast, RPE-facilitated remapping is more likely to occur when executing an action planning sequence associated with recent negative reward prediction errors, possibly due to environmental changes, and promotes exploration of alternative planning sequences.

“Based on the source of prediction errors, we consider two types of remapping: sensory prediction error (SPE)–driven remapping and reward prediction error (RPE)–facilitated remapping (Figure 1C). SPE-driven remapping is triggered when the mismatch between the predictive inputs from H to X and externally driven sensory inputs exceeds a threshold (see Materials and Methods), causing X to either transition to a different contextual state or form a new one (Figure 1D). RPE-facilitated remapping is more likely to be triggered when the agents execute an action plan following a hippocampal sequence marked by a no-good indicator. The no-good indicator indicates that the action plan, i.e. the hippocampal sequence, has recently been associated with negative reward prediction errors, possibly due to environmental changes (see Materials and Methods). It then facilitates the exploration of alternative hippocampal sequences (Figure 1E).”

In addition, we added Figure 2C-E to clarify the neural representations of external stimuli and contextual states in the X module, as well as the neural representations within the H module. We also clarified the purpose of each model component and discussed plausible biological implementations to justify our modeling choices. Furthermore, we added a schematic illustration of our results related to psychiatric disorders in Figure 5B and revised the corresponding section of the manuscript to explicitly frame these results as a computational hypothesis. We also expanded the discussion to relate our findings to existing computational psychiatry models (see point-bypoint responses below).

We believe that these revisions have improved the clarity of our model and broadened its accessibility to a wider audience.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *Ito and Toyoizumi present a computational model of context-dependent action selection. They propose a "hippocampus" network that learns sequences based on which the agent chooses actions. The hippocampus network receives both stimulus and context information from an attractor network that learns new contexts based on experience. The model is consistent with a variety of experiments, both from the rodent and the human literature, such as splitter cells, lap cells, and the dependence of sequence*

> *expression on behavioral statistics. Moreover, the authors suggest that psychiatric disorders can be interpreted in terms of over-/under-representation of context information.*

We appreciate it for carefully reading the manuscript and finding the novelty and significance in our work.

> *Strengths:*
>
> *This ambitious work links diverse physiological and behavioral findings into a self-organizing neural network framework. All functional aspects of the network arise from plastic synaptic connections: Sequences, contexts, and action selection. The model also nicely links ideas from reinforcement learning to neuronally interpretable mechanisms, e.g., learning a value function from hippocampal activity.*
>
> *Weaknesses:*
>
> *The presentation, particularly of the methodological aspects, needs to be majorly improved. Judgment of generality and plausibility of the results is hampered, but is essential, particularly for the conclusions related to psychiatric disorders. In its present form, it is unclear whether the claims and conclusions made are justified. Also, the lack of clarity strongly reduces the impact of the work in the larger field.*

We appreciate the reviewer's valuable feedback. In the revised manuscript, we have improved the presentation of the methodological aspects by providing a more intuitive and general explanation of the model framework and training procedure. We also rewrote the section on psychiatric implications to more clearly explain how dysfunction in contextual inference occurs in our model. These revisions enhance both the clarity and plausibility of our conclusions.

> *More specifically:*
>
> *(1) The methods section is impenetrable. The specific adaptations of the model to the individual use cases of the model, as well as the posthoc analyses of the simulations, did not become clear. Important concepts are only defined in passing and used before they are introduced. The authors may consider a more rigorous mathematical reporting style. They also may consider making the methods part self-contained and moving it in front of the results part.*

Thank you for raising the important point.

To improve readability, we have updated Figure 1 to more clearly illustrate the main model structure and its adaptation to individual use cases. Additionally, we have moved the previous Figure 6 (now Figure S1) to an earlier point in the Results to facilitate understanding of the methodological flow. Method section is also revised to explain the algorithmic structure indicated in Figure S1. These revisions make the methods more self-contained and easier to follow.

In the revised manuscript, we have clarified that our model is qualitatively related to the Bayesadaptive reinforcement learning framework (Guez et al., 2013) as follows.

"In the framework of reinforcement learning, our model can be mapped onto a Bayesian-adaptive model-based architecture in which contextual state serves as the root of Monte Carlo tree search (Guez et al., 2013) in a simple, largely stable environment with noiseless and unambiguous sensory stimuli, and only occasional abrupt changes. In this setup, prediction errors arise from agent's lack of experience or due to abrupt environmental changes. Once a context selector X infer the hidden state, the sequence composer H generates episodic sequences that correspond to trajectories in a search tree, each branch representing possible action–outcome sequences. Just as Monte Carlo tree search explores potential future paths to evaluate expected rewards, H produces hippocampal sequences that simulate future states and rewards based on its learned connectivity. In this way, X defines the context that anchors the root of the tree, while H expands the tree through replay or planning, thereby our model provides a simplified algorithmic implementation model-based reinforcement learning via tree search planning."

> *(2) The description of results in the main text remains on a very abstract level. The authors may consider showing more simulated neural activity. It remains vague how the different stimuli and contexts are represented in the network. Particularly, the simulations and related statistical analyses underlying the paradigms in Figure 4 are incompletely described.*

Thank you for pointing this out.

In the revised manuscript, we have added explicit examples of simulated neural activity. Specifically, we added new figures in Figure 2C–E and showed representative activity patterns from both Context selector (X) and Sequence composer (H). We also clarified the distinction between activity in the stimulus domain (externally driven) and the context domain (internally inferred states)

"Figure 2C illustrates an example of both the environmental state transition and the corresponding contextual state transition of an agent. The neural activity of X at each contextual state is shown in Figure 2D, where the environmental states … are represented in the stimulus domain and the contextual states … are represented in the context domain. … In the example transition shown in Figure 2C, the agent selected an environmental state transition from S2 to S4 in the 2nd, 5th, and 8th trials, which corresponds to a contextual state transition from X2β to X4β in the X module. However, because this transition was not rewarded, no synaptic potentiation occurred among hippocampal neurons. Subsequently, in the 11th trial, the agent attempted an environmental state transition from S2 to S5, corresponding to the transition from X2β to X5β in the contextual states.

The agent received a reward at S5, and the corresponding hippocampal sequence was strengthened, enabling the agent to acquire the alternation task in the following trials (Figure 2E)."

(see point-by-point responses below).

We also added a detailed explanation of our results in Figure 4 as follows.

"We consider a simplified environment of a probabilistic cueing paradigm (Ekman et al., 2022). In this study, two auditory contextual cues probabilistically predicted distinct visual motion sequences, and fMRI decoding was used to examine the frequency of hippocampal replay. We simplified this task as shown in Figure 4A. "

"… This result replicates Ekman et al. (2022), who showed that the probability of the contextual cues is reflected in the statistically significant differences in hippocampal replay probability in humans (Figure 4F)."

"F, Our model behavior is similar to the human fMRI result of the cue-probability-dependent hippocampal replay (Ekman et al., 2022). Paired sample t-test. **P<0.01."

We believe that these revisions make the model description and simulation results more concrete and easier to interpret.

> *(3) The literature review can be improved (laid out in the specific recommendations).*

Thank you for pointing this out. We revised the literature review to the best of our ability.

> *(4) Given the large range of experimental phenomenology addressed by the manuscript, it would be helpful to add a Discussion paragraph on how much the results from mice and humans can be integrated, particularly regarding the nature of the context selection network.*

Thank you for your suggestion.

In the revised manuscript, we added a new paragraph in the Discussion explicitly addressing how results from mice and humans can be integrated.

"Our model is a functionally modular account of the cortical regions and hippocampus, enabling it to capture experimental findings across species. While hippocampal activity in rodents has been extensively characterized in terms of spatial coding, human hippocampal representations are more often non-spatial and episodic-like (Bellmund et al., 2018; Eichenbaum, 2017). For episodic memory to support flexible behavior, it would be beneficial to retrieve each episode in a contextdependent manner. The episodic contents may vary across species and individuals, yet the fundamental computations—estimating the current context from external stimuli and their history, and flexibly updating this estimate via prediction errors—are likely conserved. Holding context information until the contextual prediction error is detected is analogous to the belief state in model-based reinforcement learning, which is known to improve performance under partially observable conditions (POMDPs) (Kaelbling et al., 1998). Our model provides a simple algorithmic implementation of this principle."

> *(5) As a minor point, the hippocampus is pretty much treated as a premotor network. Also, a Discussion paragraph would be helpful.*

Thank you for pointing this out.

We define action as a transition from one environmental state to another, and transition-coding hippocampal neurons are used for action-planning. Because our model does not incorporate errors in transitions (actions), the generated hippocampal sequences are perfectly correlated with the executed transitions (actions). However, we acknowledge that computations in the brain are more complex, with contributions from other regions such as the premotor network and the basal ganglia. To clarify this, we added formal representations of state transitions (action) in each task and the following sentences to the manuscript.

"In Sequence composer, there exist two types of neurons: state-coding neurons, which represent each contextual state, and transition-coding neurons, which encode transitions to successive contextual states given the contextual state indicated by the state-coding neurons (Materials and Methods). Note that in the real brain, not only hippocampus but also the premotor cortex and the basal ganglia contribute to action planning and execution (Hikosaka et al., 2002). Here, however, we focus on how simplified planning sequences are learned and composed in a context-dependent manner."

"Our model posits that the Sequence Composer corresponds to computations within the hippocampus. As a biologically plausible projection, we consider CA3–CA1 circuit, where contextual inputs from regions such as the PFC and EC provide the current contextual state to CA3, enabling the recurrent CA3–CA1 architecture to generate predictions of the next contextual state without errors in action."

> ***Reviewer #3 (Public review):***
>
> *Summary:*
>
> *This paper develops a model to account for flexible and context-dependent behaviors, such as where the same input must generate different responses or representations*

> *depending on context. The approach is anchored in the hippocampal place cell literature. The model consists of a module X, which represents context, and a module H (hippocampus), which generates "sequences". X is a binary attractor RNN, and H appears to be a discrete binary network, which is called recurrent but seems to operate primarily in a feedforward mode. H has two types of units (those that are directly activated by context, and transition/sequence units). An input from X drives a winner-take-all activation of a single unit H_context unit, which can trigger a sequence in the H_transition units. When a new/unpredicted context arises, a new stable context in X is generated, which in turn can trigger a new sequence in H. The authors use this model to account for some experimental findings, and on a more speculative note, propose to capture key aspects of contextual processing associated with schizophrenia and autism.*

We thank the reviewer for this summary of our model.

We would like to clarify that the hippocampal Sequence composer (H) is a recurrent network that iteratively composes the next state and the associated sensory stimuli in the sequence based on the current contextual state.

> *Strengths:*
>
> *Context-dependency is an important problem. And for this reason, there are many papers that address context-dependency - some of this work is cited. To the best of my knowledge, the approach of using an attractor network to represent and detect changes in context is novel and potentially valuable.*
>
> *Weaknesses:*
>
> *The paper would be stronger, however, if it were implemented in a more biologically plausible manner - e.g., in continuous rather than discrete time. Additionally, not enough information is provided to properly evaluate the paper, and most of the time, the network is treated as a black box, and we are not shown how the computations are actually being performed.*

We thank the reviewer for suggesting an important direction for future work. The goal of this research is to develop a minimal, functionally modular neural circuit model that provides general insights into how context-dependent behavior can be realized across species, including humans. To simplify our model, we only considered discrete-time environmental states, where the exact length of the time step depends on each environment. Extending the model to a more biologically plausible, continuous-time framework is a promising direction for future work, such as using continuous-time modern Hopfield networks and synfire chains. We modified the Discussion section to clearly point out this direction.

"... the resolution at which our model should distinguish different contextual states, including the stimulus resolution and time resolution, is hand-tuned in this work. While we used an abstract, gridlike state space with discrete time, an important direction for future work is to model its activity at finer-grained neural timescales, ... In realistic, continuously changing environments, such resolutions should be adjusted autonomously. Introducing continuous and hierarchical representations with multiple levels of spatial and temporal resolution would facilitate such adjustments, potentially through mechanisms such as modern Hopfield networks (Kurotov and Hopfield, 2020) or synfire-chain–based hippocampal sequence generation (Abeles, 1982; Diesmann et al., 1999; Shimizu and Toyoizumi, 2025; Toyoizumi, 2012), but this is beyond the focus of the current study"

Also, we would like to emphasize that our model is not treated as a black box. To improve the understandability, we have majorly revised Figures 1 and 2 to include additional details illustrating the neural activity and the internal computational mechanisms.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Major comments and suggestions for improvement:

(1) Formal link to model based RL is unclear: a core feature of inference is the role of uncertainty in modulating computation and corresponding circuit dynamics, in particular defining expected and unexpected degree of errors; as far as I understand the degree of tolerable errors within a context is defined by the size of the basin of attraction of the context module (which is dependent on number of items and the structure of correlations across patterns) and in no obvious way affected by sensory uncertainty (unless the inputs from H serve that purpose in a more indirect way). Similarly, most experiments are deemed to have deterministic (unambiguous) maps between sensory inputs and world state (although how the agent's state relates to environmental state is more complex and not completely clear based on the existing text).

Thank you for raising this important point. Our model bears conceptual similarities to model-based RL frameworks, for example, the optimal-inference formulation that underlies Monte Carlo Tree Search (Guez et al., 2013), as we now clarify in the revised manuscript. These similarities, however, are qualitative rather than quantitative. In particular, the error thresholds that separate expected from unexpected outcomes are manually specified in our model, but their exact values do not appreciably influence the simulation results.

Concretely, the heuristic threshold for SPE-driven remapping (θ_{remap}) is set to 5 bits, allowing for small miss-convergence during recall in the Amari–Hopfield model. For RPE-facilitated remapping, the threshold is set to θ_{NG} = 0.7, making the agent sufficiently sensitive to abrupt environmental changes and enabling it to explore some candidate contexts after RPE-facilitated remapping. This simple thresholding scheme is adequate for our largely deterministic simulation setting, where contextual switches are rare and occur abruptly in an otherwise stable and unambiguous environment.

Importantly, our goal in this work was not to achieve Bayesian optimality. Mice and likely humans in certain settings often deviate from optimal inference. Instead, we focus on the qualitative remapping-related processes that support goal-directed planning following epistemic errors. We have clarified this scope in the revised manuscript.

"In the framework of reinforcement learning, our model can be mapped onto a Bayesian-adaptive model-based architecture in which contextual state serves as the root of Monte Carlo tree search (Guez et al., 2013) in a simple, largely stable environment with noiseless and unambiguous sensory stimuli, and only occasional abrupt changes. In this setup, prediction errors arise from the agent's lack of experience or due to abrupt environmental changes. … However, these conceptual similarities are qualitative rather than quantitative. The goal of this work is not to achieve Bayesian optimality, but rather to show qualitative remapping-related processes that support goal-directed planning following epistemic errors."

"Note that we set the remapping threshold θ_{remap} = 5 bits to allow for small miss-convergence during recall in the Amari–Hopfield model."

"Note that we set θ_{NG} as 0.7 to make the agents sufficiently sensitive to abrupt environmental changes and enable exploring some candidate contexts after RPE-facilitated remapping."

(2) Improvement: start describing each task specification in explicit model-based RL terms, then explain how the environmental specification translates into agent

> *operations. Be explicit about what about the process is inferential, in particular, sources of uncertainty.*

Thank you for this important suggestion. Following your recommendation, we revised the manuscript to describe each task explicitly in model-based RL terms. For each task, we now identify the relevant sources of uncertainty, which arise either from imperfections in the agent's internal model of the environment or from occasional abrupt switches in task rules. We also explain how the agent infers the hidden state from experience to construct an appropriate context representation, enabling the model to perform the task successfully.

> *(3) A lot of seemingly arbitrary model choices need additional computational and biological justification; the description of the process is fundamentally an algorithmic one, which includes a lot of if-then type of operations: the dynamics of different elements of the circuit switch between "initialization to landmark/other", "error detected/not", different forms of plasticity on/off etc and it is not discussed in way how this kind of global coordination of different processes is supposed to be orchestrated biologically; e.g. as far as I understand the sequential structure in H activity is largely hardcoded rather than an emergent property of the learning+neural dynamics.*

Thank you for this important suggestion. We have made a concerted effort to clearly describe the biological context and the relevant literature motivating each of our algorithmic assumptions. Notably, as highlighted in Fig. 1F, we emphasize that the sequential structure in H activity emerges as a consequence of the agent's exploration and learning. We also explain how the two remapping mechanisms concatenate sequence segments to support long-term planning and to predict both stimuli and rewards.

About Fig. 1F

"At the beginning of learning, hippocampal segments are not connected, and H yields only short sequences that generate immediate actions and short-term predictions. As learning continues, the three-factor Hebbian plasticity rule concatenates these segments, thereby creating longer sequences that reflect the task structure (Figure 1F)."

About "initialization to landmark/other,"

"While the history-based initialization was introduced to select contextual state based on the history input from H (episodic), the landmark-based initialization was introduced to terminate the episodes that would otherwise continue indefinitely. Biologically, the landmark-based initialization corresponds to the operation of anchoring a contextual state to salient environmental landmarks - such as an animal's nest - that serve as clear reference points."

About "error detected/not,"

"Based on the source of prediction errors, we consider two types of remapping: sensory prediction error (SPE)-driven remapping and reward prediction error (RPE)-facilitated remapping (Figure 1C). SPE-driven remapping is triggered when the mismatch between the predictive inputs from H to X and externally driven sensory inputs exceeds a threshold (see Materials and Methods), causing X to either transition to a different contextual state or form a new one (Figure 1D). RPE-facilitated remapping is more likely to be triggered when the agents execute an action plan following a hippocampal sequence marked by a no-good indicator. The no-good indicator indicates that the action plan, i.e. the hippocampal sequence, has recently been associated with negative reward prediction errors, possibly due to environmental changes (see Materials and Methods). It then facilitates the exploration of alternative hippocampal sequences (Figure 1E). "

About "different forms of plasticity on/off"

“We used different learning rules for the intra-hippocampal synaptic weights depending on withinepisodic and between-episodic segments.”

“Within-episodic connections, i.e., state-coding to transition-coding synapses, are constantly updated in a reward-independent manner … This modeling is inspired by behavioral time scale plasticity in the hippocampus (Bittner et al., 2017), in which synaptic potentiation occurs for events that are close in time regardless of reward, and such plasticity is believed to support the formation of place cells, etc..”

“Between-episodic connections, i.e., transition-coding to state-coding synapses, are constantly updated in a reward-dependent manner … This is supported by the finding that dopaminergic neuromodulation gates LTP, enabling preferential consolidation of reward-associated experiences (Lisman and Grace, 2005; Takeuchi et al., 2016).”

> *(4) Improvement: Justify individual design choices by biology whenever possible; in the absence of such justification, provide at least a computational rationale for each such model choice. Additional justification for the neural substrate of different prediction errors.*

Thank you for pointing this out. Following the advice, we have added the computational objectives behind each algorithmic component in addition to the biological motivations described above. In particular, we have completely updated Fig. 1 to help readers better understand the key remapping mechanisms in our algorithm: SPE-driven and RPE-facilitated remapping.

About the Amari-Hopfield model

“We employ the Amari–Hopfield model because it allows multiple contexts to be stably maintained and selected in response to stimuli and can be trained via Hebbian plasticity. We assume that similar computations are carried out in prefrontal and entorhinal cortical circuits in the brain.” “As one possible biological implementation, we consider that Context selection in X as the brainwide evoked potential during which bottom-up information may be integrated with top-down signals to select the current context (Mohanty et al., 2025). In this case, it takes several hundred milliseconds for the contextual states in X to settle (Massimini et al., 2005).”

About the default matrix

“This contextual state is set as a default context, ensuring that the X module assigns a unique contextual state to each environmental state. Biologically, one possible interpretation is that this default context corresponds to modality-specific innate representations in prefrontal regions (Manita et al., 2015).”

About state-coding neurons and transition-coding neurons

“The state-coding neurons receive input from X and represent the current contextual state, while the transition-coding neurons send output to X and predict the next contextual state after an action ... One possible biological grounding for this functional separation is that entorhinal cortex provide contextual inputs to CA3, and CA3 and CA1 generates predictions of next state through its recurrent architecture (Chen et al., 2024).”

About the no-good indicator

“No-good indicator is introduced to transiently suppress previously established sequences that have not been recently rewarded, without devaluing them. This no-good indicator facilitates RPEfacilitated remapping (see RPE-facilitated remapping section) that leads to exploration of different contextual states in X and sequences in H. The no-good indicator is inspired by recent findings in the ventral hippocampus, where dopamine D2-expressing neurons of the ventral subiculum selectively promote exploration under anxiogenic contexts (Godino et al., 2025).”

> *(5) In particular, the temporal scale at which processes unfold with reference to behavioral time scale actions is fundamentally unclear: what determines the time scale of a sequential element? What stitches them together? What is the temporal relationship between H and X operations? At what time scale do actions happen in terms of those operating scales? How does this align with what is known about hippocampal dynamics during behavior?*
>
> *(6) Improvement: make the time scales of different aspects of the process explicit in the text, potentially with additional graphic support.*

Thank you for the questions and suggestions. In this work, we model the agent’s behavior in an abstract grid-world environment with discrete time steps, as is common in classical RL. At each time step, the agent observes a sensory stimulus, makes a plan, and executes an action based on it. The action induces a state transition in the environment. Accordingly, the model includes a single fundamental timescale: the environmental (behavioral) time step.

The modeled brain dynamics in both X and H are similarly locked to this environmental clock. As clarified in Fig. 1F, each sequence segment corresponds to one behavioral time step. These segments are then chunked based on reward events, enabling longer-horizon planning and prediction.

The agent’s cognitive operations at each behavioral time step are summarized in Fig. S1. Briefly, the agent infers the contextual state X from the current stimulus and its stimulus history, generates a sequential action plan H with predictions using chunked sequence segments, and then follows the plan when it is sufficiently promising. In addition, when sensory or reward prediction errors occur, the agent reorganizes the synaptic-weight parameters of the context selector and sequence composer. Once the agent becomes familiar with the environment, H typically generates an extended action sequence along with predictions of future stimuli and the resulting reward. The agent then executes this sequential plan, bypassing step-by-step context estimation by X, until a prediction error triggers remapping.

The revised manuscript includes the following additions.

“For simplicity, the environment is defined in discrete time, and agents move through environmental states characterized by distinct external stimuli. The model operation relies on the environmental (behavioral) time step. At each time step, the agents perform contextual state estimation by Context selector and activate a corresponding hippocampal neuron. Then, this hippocampal neuron initiates sequential activity based on hippocampal synaptic connectivity. Each hippocampal sequence represents a planned course of action and is used to predict a series of external stimuli. … The hippocampal sequence from which actions are generated is updated upon a reward. After the action execution, the agents repeat the process by selecting the current contextual state. As the agents become familiar with the environment, hippocampal sequences that enable future predictions to become longer, and contextual state estimation by Context selector becomes less frequent. The algorithmic flow chart of our model is described in Figure S1.”

> *(7) As far as I understand it, the existence of splitter cells is directly inherited from the task specification, and to some extent the same can be said about the lap cells; please explain what can be understood from the model simulations that goes beyond what was put into the inputs/reward function for each experiment. Emphasize numerical results that are counterintuitive or where additional predictions about the dynamics come directly from simulating the model but would have been less obvious beforehand.*

The existence of splitter cells in our model is not inherited from the task specification. Instead, it emerges directly from the hippocampal module retaining sensory history (namely, whether the agent approached from the left or right arm), independent of reward structure or other task details. When sensory history is removed from the sequence composer (and, consequently, from the context selector), splitter-cell representations disappear.

To develop lap-cell representations, immediate sensory history alone is not sufficient. The sequence composer must chunk episodic segments based on rewards to support sufficiently long action plans (i.e., history dependence) that span the multiple laps required by the task. The planning horizon - the length of action sequences - typically increases as animals learn a task. This progressive development of hippocampal sequences and their dependence on reward yields experimentally testable predictions. Notably, as we clarified in Fig. S2, the required sensory history length must also be learned adaptively: if it is too short, the agent cannot solve the task, whereas if it is too long, learning becomes unnecessarily slow.

In the revised manuscript, we explicitly described the emergent process of splitter cells and lap cells as follows.

About splitter cells

“A second contextual state at S2, X2β, was generated through SPE-driven remapping at the second visit of S2 (second trial) due to history mismatch… In our model, the transition-coding neurons exhibit right/left turn-specific firing at S2 after learning is complete (Figure 2E, I), replicating the emergence of splitter cells.”

About lap cells

“the task environment changes again and the agents are rewarded for two laps, …. Either the shortest transition, ..., or the one-lap transition, …, is no longer rewarded, which triggers another RPE-facilitated remapping and exploration. During exploration, a history mismatch occurs …, and the contextual states for the second lap … are generated. Finally, the rewarded transition of contextual states and corresponding sequence… is reinforced (Figure 3B).”

“This task can also be solved by simply preparing temporal contexts with three steps of sensory history (n=3), which is the minimal number to solve this task. (see Materials and Methods for Model-free learning). However, it takes much longer to find the correct transition for solving the 1-lap task than our model because it involves an excessive number of states (Figure S2).”

“As the agents become familiar with the environment, hippocampal sequences that enable future predictions to become longer, and contextual state estimation by Context selector becomes less frequent.”

> *(8) The partitioning of H subpopulation into current input vs predictive subpopulations seems to fundamentally deviate from known CA1 properties like theta phase processing, where the same neurons encode information about recent past, present, and future at different moments in time within a theta cycle. The existence of such populations (especially since they come with distinct plasticity mechanisms and projection patterns) seems like a strong avenue for validating the model experimentally.*

> *(9) Improvement: biologically justify the two subpopulations, discuss neural signatures of this distinction that could be used to identify such neurons in experiments*

We thank the reviewer for bridging our model with biological circuits.

First, we would like to clarify that we do not claim that our H module corresponds to CA1 specifically.

Rather, we assume that within the broader hippocampal loop (EC–DG–CA3–CA1–EC), subpopulations emerge that preferentially encode the current contextual states and the transitions to the next contextual states. This assumption reflects our hypothesis that the hippocampus implements a mechanism for predicting the next context given the current one. Importantly, this functional separation does not contradict known theta-phase coding in which the same neurons can represent past, present, and future information at different phases of the theta cycle.

As a possible biological grounding, we particularly emphasize the CA3–CA1 projection. Recent studies have shown that CA1 representations exhibit a temporal delay relative to CA3 activity (Chen et al., 2024), suggesting a circuit-level mechanism by which predictions of upcoming contextual states may be computed based on the current context. In this framework, state-coding and transition-coding functions could be assigned to CA3 and CA1, or dynamically expressed through their interactions. Based on our model, we make testable experimental predictions. Specifically, we predict that neural representations in CA3 and CA1 should precede contextual switching in tasks such as alternation or multiple-lap tasks, and that perturbing CA3–CA1 computations would impair task performance.

Please note, however, that our model does not characterize the sequence composer's activity at such fine-grained neuronal timescales. Instead, we model the computation it performs in abstract time steps corresponding to the grid states (e.g., while the animal is at a corner of the maze).

We have added these points to the Discussion to clarify the biological interpretation and to suggest potential experimental validations of the proposed subpopulation distinction as follows.

"Our model posits that the Sequence composer corresponds to computations within the hippocampus. As a biologically plausible projection, we consider the CA3–CA1 circuit, where contextual inputs from regions such as the PFC and EC provide the current contextual state to CA3, enabling the recurrent CA3–CA1 architecture to generate predictions of the next contextual state. Consistent with this idea, the temporal lag in CA3→CA1 transmission suggests a functional gradient in which CA3 represents present-oriented information while CA1 carries more futureoriented predictions (Chen et al., 2024), and neurons in both CA3 and CA1 exhibit action-driven remapping and encode action-planning signals (Green et al., 2022). Our framework, therefore, predicts that changes in CA3→CA1 population activity precede behavioral switching in contextdependent alternation in Figure 2 or multi-lap tasks in Figure 3, and perturbation of this input will degrade the behavioral performance."

"While we used an abstract, grid-like state space with discrete time, an important direction for future work is to model its activity at finer-grained neural timescales, such as theta cycles (Foster and Wilson, 2007; Wikenheiser and Redish, 2015)."

> *(10) The flexibility of the new solution in terms of learning contexts with variable temporal horizons seems an important feature of the model, but one poorly demonstrated in the existing numerical experiments. Could more concrete model predictions be generated by designing an experiment targeted specifically for such scenarios?*

Thank you for raising this point.

As we showed in Figure S2, in environments with variable temporal horizons, our model performs better than model-free learning (Q-learning) that incorporates temporal context.

To further demonstrate this point, we added a new task in Figures 3G and H, in which the 1-lap task and the 2+ lap task are alternated. Our model exhibits rapid switching between these tasks, regardless of differences in sequence length or temporal horizon. We added the following text.

“To demonstrate the advantage of our model in a rapidly switching task that requires different history lengths, we show that an agent trained on both the 1-lap and 2-lap tasks can flexibly alternate between them in a reward-dependent manner (Figure 3G), selectively engaging hippocampal sequences of different lengths according to the current task context (Figure 3H). Together, these results illustrate how hippocampal lap-like representations emerge through learning and enable flexible context switching across tasks with distinct temporal demands.”

In such a scenario, a subjective representation of laps in the hippocampus is the key to solving the task. As we responded to points (8) and (9), neural representations, especially in CA1, are expected to bifurcate between the 1-lap and 2-lap conditions, and this bifurcation would precede and critically govern the animal’s behavior.

> *(11) I found figures confusing/uninformative, specifically in making it explicit what is external task structure and what is the agent's internal representation of it; as a result it is not clear what of the results is trivially inherited from the task specification and what is an emergent property of the model; e.g. Figure 2A described external transition specification according to world model but it is unclear to me if Figure 2B shows the ideal agent state representation across context or a graphical summary of what the agent actually learned from the sensory experience described in A; from the text. Figure 2F is supposed to describe a property of the emergent representation, but what is shown is another cartoon... etc.*

We appreciate the reviewer’s insightful comments regarding the clarity of our figures.

To clarify the neural representation of the agent and how it links to the action, we have revised Figure 2 and the descriptions in the main text.

First, Figure 2A schematically depicts the external stimulus as being determined solely by the task. In this task, animals must keep track of the immediately preceding state (S1 or S3) to correctly choose between S4 and S5 upon reaching S2. Without such a memory of prior states, an agent would have no basis for distinguishing which action is appropriate, and therefore cannot selectively move to S4 and S5. Therefore, any reinforcement learning model that does not incorporate at least a onestep state history cannot solve the task.

To solve the task, S2 must be represented as two distinct contextual states depending on the previous state. Figure 2B therefore illustrates an example of internal representation that separates S2 into X2α and X2β: transitions from S1 to S2 are internally represented as X1 → X2α, whereas transitions from S3 to S2 are represented as X3 → X2β. Although the sensory inputs provided to the model correspond only to the task-defined states in Figure 2A, the combination of the sensory input with contextual states in Context selector successfully achieves this contextual representation of X2α and X2β (see Figure 2C, D). Also, the hippocampal neurons in Sequence composer indicate the next contextual states given the current contextual states, i.e., X2α→X4 and X2β→X5 (see Figure 2E). Thus, combining Context selector and Sequence composer successfully achieves the task requirement indicated in Figure 2B.

Regarding the reviewer's concern that Figure 2F (now Figure 2I) appeared to be another cartoon, we have revised the panel to clearly display our result. These results demonstrate that some hippocampal neurons in our model encode the transition from X2α→X4 and X2β→X5. The updated figure clarifies that our hippocampal neurons functionally work similarly to the splitter cells in Wood et al., 2000.

> *(12) Improvement: use visuals and captions. Make it clear what is a cartoon, what is a model specification, and what is an actual result. Replace/complement algorithmic cartoons in Figure 1 with a description of the actual result.*

Thank you for raising this point.

As we explained in the previous point (11), we added Figure 2D and Figure 2E for displaying the actual neural activity, and the corresponding annotations in the manuscript, e.g, X2α. Also, we revised the cartoons of our model description in Figure 1 to better describe our model structure.

> *(13) Map between model and experimental results is poorly justified: in particular the nature of sensory inputs is not clearly specified, and how the experimental manipulations (e.g. MEC input disruption) map into model manipulations is not intuitive and no justification is provided for the choices beyond that the model ends up matching the experiment by some metric. Also, not clear why a tradeoff of neural resources as implemented in the model makes sense for the clinical case and how this hypothesis deviates from alternative Bayesian accounts invoking imperfections in inference (e.g. relative strength of priors vs likelihood as reported by e.g. P.Series's group, or issues with hierarchical inference more generally along R.Jardri's work).*

Thank you for raising this important point. We have revised the manuscript to clarify the mapping between model components, sensory inputs, and the experimental manipulations, and to further justify the clinical interpretation.

About sensory inputs

First, each environmental state in our model is represented as a binary (0/1) pattern. We have added Figure 2D to explicitly illustrate these sensory stimuli and how they are provided to the context-selection module.

About mapping between model components and brain circuits

Functionally, we speculate that Context selector (X) corresponds to computations carried out in the prefrontal cortex (PFC) and entorhinal cortex (EC), and Sequence composer (H) corresponds to the hippocampus. Inputs from the PFC are thought to reach the hippocampus via the EC. Therefore, suppression of MEC→hippocampus inputs in Sun et al. (2020) naturally maps onto blocking a subset of the inputs from X to H in our model.

We clarified this correspondence in the revised manuscript and now explicitly justify why this manipulation matches the biological experiment.

Relation to Bayesian theories

We agree that Bayesian accounts have provided influential explanations of psychiatric symptoms by invoking imperfections in inference, such as imbalances between priors and likelihoods (e.g., work by P. Series and colleagues) or disruptions in hierarchical inference (e.g., work by Jardri and others). Our model complements these frameworks by explicitly incorporating sequential structure and context remapping. Rather than treating priors as static or fixed-weight quantities, our model allows contextual representations to be dynamically reorganized based on prediction errors over time. In the SZ-like condition, we assume that an excessively expanded context domain increases the influence of internally generated contextual predictions, causing them to override sensory inputs and resulting in maladaptive behavior with hallucination-like percepts. Importantly, this effect reflects not only stronger priors but also excessive generation and competition of contextual states, leading to unstable and non-reproducible remapping. In contrast, in the ASD-like condition, sensory-weighted context representations limit the ability to flexibly incorporate newly introduced contexts, causing the model to perseverate on an initially learned context and thereby reproduce inflexible behavior. We added a schematic illustration in Figure 5B and expanded the Discussion to clarify this point.

"When the stimulus domain is relatively underrepresented, the reconstruction of contextual state in the Amari-Hopfield network tends to infer contextual states based on the context domain rather than the stimulus domain. Consequently, it converges to an incorrect attractor that is not assigned to the current environmental state, thereby increasing perceptual error for external stimuli (hallucination-like effects). Moreover, SPE-driven remapping and the corresponding synaptic plasticity occur more frequently. In contrast, when the stimulus domain is overrepresented, the Amari-Hopfield network rarely assigns multiple contextual states to a given environmental state, leading to an overuse of default contextual states (see Figure 5B and Materials and Methods). "

"Our model also provides an algorithmic-level account of psychiatric symptoms by changing the relative weighting of sensory-encoding versus context-coding neurons. This implementation is analogous to Bayesian theories linking priors to psychiatric symptoms. In SZ, hallucinations and delusions have been modeled as arising from overly strong top-down priors (Powers et al., 2016) or circular inference, which leads to erroneous belief formation (Jardri et al., 2017; Jardri and Denève, 2013). In our model, we used an underrepresented stimulus domain to increase the relative influence of internally generated context representation in context selection. Crucially, this implementation does not simply strengthen priors but induces excessive generation and competition of contextual states, leading to frequent yet non-reproducible remapping of hippocampal contextual activity and a failure of learning to converge despite repeated experience. In ASD, it has been argued that abnormally high sensory precision reduces the updating of expectations (Karvelis et al., 2018) or leads to sensory-dominant perception, which has been interpreted as weak priors (Angeletos, Chrysaitis, and Seriès, 2023; Lawson et al., 2014; Pellicano and Burr, 2012). In our framework, we used an overrepresented stimulus domain to increase the relative influence of external stimulus representations in context selection. Importantly, our model captures not only sensory-dominant processing emphasized in previous studies, but also a distinctive impairment in flexibly utilizing newly introduced contexts, reflecting a failure of context reconstruction and resulting in persistent inflexible behavior. Thus, our conjunctive modeling of sensory and context processing complements Bayesian accounts of psychiatric symptoms and provides a mechanistic explanation for the role of sensory processing in maladaptive, inflexible behavior. "

> *(14) Improvement: justify choices, explain in more detail relationships with computational psychiatry literature.*

Thank you for pointing it out. As we explained in the previous point (13), we justified our model choice in the revised version.

> *Minor comments:*
>
> *(1) Typos: "algorism" (pg2), duplicate Sun reference.*

Thank you for finding the typo and the missing reference. We revised accordingly.

> *(2) Unclear statements from Methods:*

> - *"preparing temporal context with three histories" not sure what is meant by this.*
> - *"... state estimation by the context-selection module becomes less frequent." (Methods/Overview): what is the mechanism?*
> - *"default pattern" and failure to converge: What is the biological basis for them?*
> - *Why is the converter function used on some occasions but not others?*
> - *"new contextual state is prepared": What does that mean?*

We thank the reviewer for pointing out several unclear statements in the Methods section.

- “preparing temporal context with three histories”

We now explicitly state the formal description of three histories in the Methods as follows.

“the state is defined by the recent n-step transition history of task state (i.e. ${s_k}^{(n)}$ =(S_k,S_{k-1}, ⋯,S_{k-n})^T , where ${s_k}^{(n)}$ is the temporal context state, and S_k is the environmental state at time k). We changed n from 0 to 3.”

- “state estimation by the context-selection module becomes less frequent”

In our model, context selection is performed every time the agents execute an action sequence generated by Sequence composer. As learning progresses, the Sequence composer comes to predict distant future states and executes coherent action sequences based on these predictions. When no unexpected errors are encountered during execution, context estimation is suppressed, resulting in less frequent context selection. We modified the manuscript as follows.

“After the action execution, the agents repeat the process by selecting the current contextual state. As the agents become familiar with the environment, hippocampal sequences that enable future predictions to become longer, and contextual state estimation by Context selector becomes less frequent. The algorithmic flow chart of our model is described in Figure S1.”

- “default pattern”

In biological systems, it is reported that the frontal cortex shows sensory modality-specific representation without prior learning (Manita et al., 2015). We refer to these innate modalityspecific sensory representations as the default pattern. In the early stages of learning, we assume that no stable contextual representations have yet been formed in the brain, and therefore, a default pattern uniquely driven by external stimuli is used as the context representation. Even during intermediate stages of learning, the context selector may fail to converge to a specific state. In such context-uncertain environments, it has been reported that agents often rely on previously learned or habitual action choices (psychological inertia), which is evident in ASD patients.

“This contextual state is set as a default context, ensuring that the X module assigns a unique contextual state to each environmental state. Biologically, one possible interpretation is that this default context corresponds to modality-specific innate representations in prefrontal regions (Manita et al., 2015).”

“This default implementation is analogous to psychological inertia, particularly under uncertainty (Ip and Nei, 2025; Sautua, 2017), which has been reported to be more pronounced in ASD patients (Joyce et al., 2017).”

- Why is the converter function used only in some cases?

The converter function A(stim → context) was introduced to compose the default pattern (one-toone mappings between stimuli and contexts) as we described above. In other cases, the Hopfield dynamics were used to select contextual states; therefore, we did not use the converter function.

- “new contextual state is prepared”

Thank you for pointing this out.

The term “prepared” was inaccurate. We revised it to “generated”.

In the case of remapping, we assumed that X generates a new random neural activity pattern in its contextual domain and stores it as a new contextual state. We described this process as “a new contextual state is generated”.

> *(3) Please explain the mapping between hippocampal sequences to actions in more detail for each task.*
>
> - *Why 9 attempts before rejection?*
> - *Why all the variations on Hebb?*

We appreciate the reviewer’s request for clarification. Below, we provide additional explanations point by point.

Mapping between hippocampal sequences and actions

In this research, we defined action as the transition from one environmental state to another environmental state. The hippocampal sequences predict the transition of environmental states; therefore, they correspond to a set of action plans from the current environmental state. In the revised manuscript, we added the formal definition of environmental states and actions in each task.

- Why 9 attempts before rejection?

These repetitions ensure adequate exploration of the contextual states in X and the episodic sequence in H before committing to an action. Increasing the number of attempts excessively causes the reward value function $(V_{\tilde{H}_{-1}})$ to be dominated by a single highest-scoring sequence, thereby causing excessive exploitation and narrowing behavioral variability. While the exact number 9 is not critical—the qualitative results are robust to moderate changes—we selected this value because it provides a good balance between exploration and exploitation and produces the clearest visualizations in our figures. We have clarified this in Method below.

“We set the number of attempts before rejection to nine, providing a balance between exploration and exploitation and serving as a good compromise for visualization.”

- Why all the variations on Hebbian learning?

We consider three loci of plasticity in our model: the X module, the H module, and their reciprocal connections. Within the H module, synaptic connections that link episodic segments—specifically from transition-coding neurons to state-coding neurons—are assumed to follow a reward prediction error–dependent, supervised form of Hebbian learning. This choice reflects the need to selectively reinforce transitions that lead to successful outcomes. In contrast, all other synaptic updates in the model are assumed to follow reward-independent, activity-based Hebbian learning. These learning rules support the unsupervised formation and stabilization of contextual representations and action execution.

In addition to the basic Hebbian rule, we introduced biologically motivated constraints, such as upper and lower bounds on synaptic weights and heterosynaptic depression, which weakens nonpotentiated synapses. Importantly, these mechanisms do not alter the fundamental nature of Hebbian learning but increase the stability of our model.

> *(4) For Q learning: please clarify "the state is defined by the recent transition history of task state.*

As you suggested, we clarified the statement by adding the following sentences in Method. “To highlight the advantage of our model, we compared it to the Q-learning with temporal contexts, namely, the state is defined by the recent n-step transition history of task states (i.e. ${s_k}^{(n)}$ =$(S_k,S_{k-1},\cdots,S_{k-n})^T$, where ${s_k}^{(n)}$ is the temporal context state, and S_k is the environmental state at time k.”

> *(5) What is the purpose and biological justification for the NG addition to RW?*

Thank you for raising this point. The prediction-error–based update of each sequence’s value function R alone cannot distinguish between two fundamentally different cases:

(a) the value of a sequence has genuinely decreased, or

(b) the sequence remains useful, but it is just not appropriate in the current context. This distinction is essential for modeling context-dependent switching of behavioral strategies. To address this, we introduced the No-good (NG) indicator. NG allows the agent to temporarily mark certain sequences as unsuitable without altering their long-term value, thereby facilitating short-term exploration of alternative sequences. In other words, NG provides a mechanism for transiently suppressing a previously valid sequence in case of contextual changes, while preserving the underlying value learned in past experiences.

This mechanism is consistent with several lines of biological evidence. First, extinction learning after fear conditioning does not erase the original fear memory but instead forms a new memory trace, known to be stored in the medial PFC (Milad & Quirk, 2002). This suggests that animals may switch to a different contextual representation rather than simply downgrading the value of the conditioned stimulus, supporting the idea of temporarily suppressing a sequence without modifying its intrinsic value.

Second, recent studies in the ventral hippocampus show that dopamine D2–expressing neurons in the ventral subiculum promote exploration specifically under anxiogenic contexts (Godino et al., 2025). This finding is consistent with the short-term exploratory behavior enabled by our NG mechanism. Thus, we added the following statement to the manuscript:

“No-good indicator is introduced to transiently suppress previously established sequences that have not been recently rewarded, without devaluing them. This no-good indicator facilitates RPEfacilitated remapping … that leads to exploration of different contextual states in X and sequences in H. The no-good indicator is inspired by recent findings in the ventral hippocampus, where dopamine D2-expressing neurons of the ventral subiculum selectively promote exploration under anxiogenic contexts (Godino et al., 2025).”

Together, these biological findings provide a conceptual basis for modeling NG as a contextsensitive, transient modulation that encourages exploration without overwriting previously learned sequence values.

> *(6) Missing details about H network size*

Thank you for pointing it out.

We used 300 neurons for H. We indicated it as below.

“We model the hippocampus with an N = 300 binary recurrent neural network.”

> *(7) S1 figure: learning is slower even in the early, easy phases of learning when the temporal dependence should not matter; how are learning rates calibrated across models?*

Thank you for raising this point. In our model, the learning rate was fixed at 0.15, whereas the control model (now shown in Figure S2) uses a higher learning rate of 0.4, independent of temporal context.

Regarding why learning appears slower even in the early, easy phases, when the number of temporal contexts increases, the size of the state space expands. This broadening of the state space makes it more time-consuming to identify and reinforce the appropriate state transitions. This is especially evident in easy phases because the temporal context prepared in the model is excessive to the number of temporal contexts that the task requires.

Importantly, unlike the control model, which postulated a fixed number of temporal contexts, our model gradually increases the number of temporal contexts depending on prediction error. This adaptive mechanism allows the model to achieve fast learning during early, easy phases while still enabling more complex learning in later phases.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) "Hippocampal neurons show sequential activity...." The authors should include more classical references for hippocampal sequential activity at this point, too.*

Thank you for your suggestion. We added the citations below

Skaggs and McNaughton, 1996; Wilson and McNaughton, 1993

> *(2) "...called remapping" also here, please reference classic work (Bostock, Muller, ...)*

As suggested, we added the citations below

Bostock et al., 1991; Muller and Kubie, 1987

> *(3) "Several theoretical models..." What I miss here are models that explain remapping by inputs from the grid cell population, and/or the LEC (see Latuske 2017 for review), still widely considered the standard mechanism. Also, the models by Stachenfeld et al. 2017, Mattar and Daw 2019, and Leibold 2020 specifically address context dependence. Accordingly, "A comprehensive model that can explain the formation of context-dependent hippocampal sequences of various lengths through remapping, while relying on a biologically plausible learning process,..." somewhat overstates the novelty of the current paper.*

Thank you for pointing this out and for suggesting relevant citations. We agree with the reviewer that inputs from MEC and LEC to the hippocampus constitute a fundamental mechanism underlying remapping. However, in our view, a key open question in the remapping field is how MEC and LEC estimate the current context and convey this information to the hippocampus in a manner that supports goal-directed behavior. While previous studies have addressed remapping at the representational level and the hippocampal sequence at planning, the overall relationship between remapping, reinforcement learning, and planning has not yet been explained within a single unified model. In this work, we propose a simple and biologically plausible model that integrates an Amari–Hopfield network for context selection with hippocampal sequences, providing an account of coordination under goal-directed behavior. To more accurately position the novelty of our contribution, we have revised the manuscript as follows.

"While previous works have explored hippocampal sequential activity for planning (Jensen et al., 2024; Mattar and Daw, 2018; Pettersen et al., 2024; Stachenfeld et al., 2017) and hippocampal remapping for contextual inference (Low et al., 2023) separately, they have yet to elucidate how these two aspects jointly enable flexible behavior. A simple biologically plausible model-based reinforcement learning model that uses the Amari-Hopfield model for context selection and hippocampal sequences of various lengths as a state-transition model for long-horizon planning, relying on remapping driven by prediction errors to form state representation, would thus provide valuable insights into the neural mechanisms underpinning context-dependent flexible behavior."

> *(4) Please properly introduce nomenclature "C2α, C2β, S2,...." S is sometimes used for stimulus, sometimes for location (state?), or even action?*

Thank you for pointing it out. We acknowledge that the annotation of Cn (e.g., C1, C2…) was not straightforward. Therefore, we changed the annotation to Xn (e.g., X1, X2, …) in order to indicate the contextual state of X.

We define Sn (e.g., S1, S2…) as the external input given by the environment and represented in stim. domain of X, while Xn (e.g., X1, X2…) is the subjective contextual state generated by the agent and represented in the context domain of X. As a reference, we added the neural representation of X in Figure 2D and added the following text below.

"The neural activity of X at each contextual state is shown in Figure 2D, where the environmental states (e.g., S1, S2…) are represented in the stimulus domain, and the contextual states (e.g., X1, X2α…) are represented in the context domain."

> *(5) "Our model replicates this result by blocking the synaptic transmission from most of the neurons in the context domain of X to H (Figure 3F).". Does this mean the X module is hypothesized to be in the EC?*

Thank you for the thoughtful question. In our model, the X module is intended as a functional abstraction that combines the roles of several brain regions known to contribute to contextual representation, including the prefrontal cortex (PFC) and the entorhinal cortex (EC). Although X is not necessarily meant to correspond to a single anatomical region, we consider it likely that the contextual information represented in X would reach the hippocampus (H) (CA3 and CA1) primarily through the EC. Thus, the experimental manipulation shown in Figure 3F—suppression of medial EC axon at the hippocampus—is interpreted in our framework as weakening the input from X to H.

We added the following texts in the Discussion section.

"We speculate that Context selector is implemented across multiple brain regions with varying degrees of resolution, including a part of the entorhinal cortex and prefrontal cortex."

"Our model posits that the Sequence Composer corresponds to computations within the hippocampus. As a biologically plausible projection, we consider the CA3–CA1 circuit, where contextual inputs from regions such as the PFC and EC provide the current contextual state to CA3, enabling the recurrent CA3–CA1 architecture to generate predictions of the next contextual state."

> *(6) Discussion "model-based reinforcement learning": Please detail where the model is here. In my understanding, the naive agent does not have a model (this would be model-free then?).*

Thank you for asking.

Unlike model-free reinforcement learning, where each action is evaluated step by step, we use hippocampal sequences for multiple-step prediction and action planning. This is the “model” in our research. As you mentioned, initially, animals do not have a “model”, but Sequence composer gradually chunks the episodic segments to compose a longer sequence.

> *(7) "...can change the attractor dynamics in the hippocampus (34)": What is (34)? I also would doubt that one can make such absolute statements about the human hippocampus.*

Thank you for pointing out the missing citation. We corrected it accordingly.

Rolls E. 2021. Attractor cortical neurodynamics, schizophrenia, and depression. Transl Psychiatry 11. doi:10.1038/s41398-021-01333-7

> *(8) "To the best of our knowledge, this is the first model that describes the formation of contextdependent hippocampal activity through remapping and its contribution to flexible behavior." See "Several theoretical models...".*

Thank you for pointing this out. We admit that it was an overstatement. We corrected it accordingly.

“To the best of our knowledge, this is the first model that uses associative memory for describing the formation and switching of context-dependent hippocampal activity through remapping and its contribution to flexible behavior.”

> *(9) "We speculate that the context-selection module is implemented across multiple brain regions..." How would an attractor network be implemented over "multiple brain regions"?*

We thank the reviewer for raising this important conceptual question. Context information in realistic environments is likely to have a hierarchical structure. We therefore speculate that multiple brain regions may jointly support context selection by maintaining different levels or components of this hierarchy. In particular, the prefrontal cortex (PFC), medial entorhinal cortex (MEC), and lateral entorhinal cortex (LEC) have all been implicated in representing contextual or task-state information at different levels of abstraction. These regions are known to exhibit attractor-like dynamics and to provide inputs to the hippocampus. Thus, an attractor network spanning multiple regions could arise, with different areas stabilizing distinct components of the contextual representation, depending on the timescale of memory, task demands, or sensory features.

We used the Amari–Hopfield network as a functional abstraction to explain such multi-regional interactions underlying context representation, rather than to provide a one-to-one mapping onto a specific brain region. How region-specific attractor dynamics jointly contribute to maintaining global contextual information and enabling context switches in response to prediction errors remains an important direction for future research.

> *Methods:*
>
> *(10) "... agents move through discrete environmental states characterized by distinct external stimuli.": How is this exactly implemented? What is the neural representation of these states, xi? What is the difference to a "landmark"?*

We appreciate the reviewer’s thoughtful question regarding the implementation and neural representation of environmental states. In our model, each environmental state is represented as a binary stimulus pattern provided to the stimulus-domain neurons in Context Selector. Specifically, for each state, we constructed a pattern in which half of the neurons are set to 1 and the other half to 0. We chose this design because, in the Amari–

Hopfield model, memory performance is maximized when stored patterns contain approximately equal proportions of 0 and 1. For clarity, we have added an illustration of these stimulus patterns in the revised Figure 2D.

Regarding the reviewer's question about landmarks: in our framework, a landmark denotes an environmental state for which the contextual state is uniquely determined, regardless of the preceding transition history. For simplicity in this study, we designated the initial environmental state in each task (S0 or S1) as the landmark. Importantly, in our implementation, landmarks do not differ from other states in terms of their stimulus pattern; their special role arises solely from the task structure, not from additional sensory properties.

In real environments, what constitutes a landmark likely varies depending on stimulus saliency and the agent's prior experience. Determining how landmarks should be optimally defined or learned is an interesting direction for future work.

> *(11) How are different contexts represented for the same stimulus xi^stim?*

We added an example of neural activity in X in Figure 2D, illustrating the distinction between the stimulus domain and the context domain. While the activity in the stimulus domain depends on the external stimulus, the contextual domain consists of uncorrelated random neural states. We exploit a key property of the Amari–Hopfield network to associate each contextual state with a given external stimulus.

> *(12) "...and its stimulus domain ??stim becomes identical to ??xistim ." Does that mean every stimulus is an attractor in the context net? How can that work with only 1200 neurons? Is that realistic for real-life environments? Neuron numbers would need to increase dramatically.*

As you mentioned, we assigned each stimulus to a corresponding attractor in the Context selector (X). An Amari–Hopfield network with 1,200 neurons can store approximately 10–20 attractors, which is sufficient to solve the tasks considered in this study. We adopted the Amari–Hopfield network for its simplicity and conceptual clarity; however, in biological neural systems, it is not necessary to construct such rigid attractors for every stimulus. For example, modality-specific neural projections exist in the brain and are sometimes sufficient to form loose attractor states across different stimuli. In addition, the prefrontal cortex is known to support working memory, which may also serve as a form of contextual representation incorporating recent history. Thus, we propose that multiple brain regions cooperate to implement the Context selector.

> *(13) How are WHX and WHH initialized?*

Thank you for pointing this out.

We set the initial condition of all W to 0. We added the following text in the Method section.

"Note that the initial synaptic weights of W^{HX} and W^{XH} are all 0."

> *(14) It is unclear why the hippocampus separates into state and transition neurons. Why cannot one pattern serve both purposes?*

Thank you for asking about this important point.

The reason why we prepare two kinds of hippocampal neurons is that state-coding neurons represent the current contextual state, and transition-coding neurons predict the following contextual state under the current contextual state. These two separations enable it to predict multiple scenarios under the current contextual state and to choose a sequence most suitable in the environment.

We rewrote the following sentences in the manuscript.

In result section,

“In Sequence composer, there exist two types of neurons: state-coding neurons, which represent each contextual state, and transition-coding neurons, which encode transitions to successive contextual states given the contextual state indicated by the state-coding neurons”

In Method section,

“The state-coding neurons receive input from X and represent the current contextual state, while the transition-coding neurons send output to X and predict the next contextual state after an action i.e., $\mathrm{T}(X_{k+1} \mid X_k, a_{k,k+1})$.”

> *(15) "the agents execute actions according to this sequence." How are the actions defined? Are they part of the state?*

We thank the reviewer for raising this important point. In our model, an action is defined as the transition from a given environmental state to the next environmental state. To avoid ambiguity, we have added a formal mathematical definition of actions for each task in the revised manuscript. In our framework, the transition-coding neurons in Sequence Composer (H) predict the upcoming environmental state, and thus the hippocampal sequence intrinsically contains the representation of an action. Consequently, the sequence generated before actions functions as the agent’s internal action planning process.

> *(16) "Because the input source for the state-coding neuron and the transition coding neuron differ (the former is selected from ??, while the latter is selected from ??), the same hippocampal neuron could occasionally be used for both state-coding and transition-coding across different contextual states. This is evident when an excessive number of contextual states are prepared, especially in the SZ condition. This phenomenon degrades state estimation at X (eq.3)." I have no idea what you want to convey here, ... and how is state estimation related to Equation 3?*

We appreciate the reviewer’s feedback and agree that our original explanation was unclear. Our intention was to clarify why context estimation deteriorates specifically in the SZ condition.

In our model, state-coding neurons in the hippocampus represent the current contextual state, and transition-coding neurons predict the next contextual state given the current contextual state. Under normal conditions, these two sets of neurons remain sufficiently distinct, allowing accurate prediction of the upcoming contextual state, which is conveyed to X. However, when an excessively large number of contextual states are stored in the SZ condition, representations in the hippocampus begin to overlap. As a result, some hippocampal neurons are inadvertently recruited for both state-coding and transition-coding across different contextual states. This overlap disrupts the H’s ability to accurately predict the next contextual state.

This degraded prediction directly affects the state-estimation process in X (Eq.3), because Eq.3 relies on receiving an accurate predicted next state from H. When this signal becomes ambiguous, X may converge to an incorrect contextual state, potentially mimicking hallucination-like inference errors.

We have rewritten the relevant passage in the manuscript to clarify this mechanism as follows.

“When the number of contextual states increases - particularly in the SZ condition - representational overlap arises between hippocampal state-coding and transition-coding neurons.

This overlap makes the prediction of the next contextual state by the transition-coding neurons unreliable. The degraded prediction from H, in turn, corrupts the initial condition for context selection in X (Eq. 3), leading to hallucination-like behavior.”

> *(17) The figures hardly show simulated activity. Consider displaying more neuronal simulations to help the reader grasp the workings of the model.*

Thank you for your suggestion. We indicated the neural activity of X and H in Figures 2D and 2E, respectively, to show the overview of our model.

> *(18) Figure 5: What is the "Hopfield count"?*

Thank you for pointing this out. The definition of the Hopfield count was ambiguous. We added an explicit explanation of “context selection” and its possible outcomes (correct association, hallucination-like, and default contexts) in Fig. S1. To clarify our claim, we replaced the countbased measure with the probability of selecting hallucination-like and default contexts during context selection. Accordingly, we removed the term “Hopfield count” and revised the caption of Figure 5 as follows.

“The result of context selection (see Figure S1). The probability of wrong stimulus reconstruction (hallucination-like effects) is plotted in red, and the probability of default context usage due to failures in context reconstruction (see Materials and Methods) is plotted in blue.”

> *(19) Figure 6: Consider moving this upfront.*

Thank you for the suggestion. We moved Fig.6 to Fig.S1 and introduced it earlier in the manuscript.

> ***Reviewer #3 (Recommendations for the authors):***
>
> *I was a bit confused about the implementation, which may not be autonomous, meaning there are numerous stages that require intervention from outside the X-H network (see Figure 6). It seems that the X network might wait to converge before providing input to H, rather than having the entire network evolve in parallel. There are also aspects to the implementation that seem rather ad hocsuch as the "no-good indicator".*

Thank you for the thoughtful comments. We would like to clarify several points regarding the implementation and its biological motivation.

First, regarding the concern that the X–H interaction may not be fully autonomous:

In our framework, the convergence time of the X module under external sensory input is assumed to be on the order of several hundred milliseconds, consistent with the timescale of stimulus-evoked cortical population dynamics observed in biological systems. Especially when hippocampal input is present, X does not need to explore the full attractor landscape. Instead, it quickly settles into an attractor located near the hippocampal cue, which substantially shortens the convergence time.

Second, although our current implementation proceeds in an algorithmically sequential manner for clarity, we do not intend to imply that the brain performs these steps sequentially. Biologically, the states of X and H are expected to co-evolve and mutually constrain each other through recurrent interactions. The sequential algorithm in the model is therefore a practical choice for implementation, not a theoretical claim about strict temporal ordering in the neural system.

Finally, the “no-good indicator” is introduced to suppress hippocampal sequences transiently and thereby accelerate switching behavior. Our no-good indicator is most consistent with the biological findings on D2-expressing neurons in the hippocampus. We added the following text below.

About the no-good indicator

“The no-good indicator is inspired by recent findings in the ventral hippocampus, where dopamine D2-expressing neurons of the ventral subiculum selectively promote exploration under anxiogenic contexts (Godino et al., 2025)”

Besides the hippocampus, similar mechanisms—temporary suppression of recently visited or lowvalue attractor states—have been proposed in biological decision-making and working-memory literature, providing conceptual support for the no-good indicator in our model.

> *After exposure to a new context, a new memory/context is stored in the X network. As the storage of a new memory requires synaptic plasticity, this step would presumably take a significant amount of time in an animal.*

Thank you for raising this important point. We agree that the formation of a new memory or context requires synaptic changes, and it is well established that processes such as tagging during wakefulness and consolidation during sleep take considerable time. However, once a context has been learned, switching between contexts can be achieved just by moving between attractors in the X network. This mechanism allows for rapid, context-dependent behavior without requiring new synaptic modifications each time. Our study focuses on this aspect of fast context-dependent switching rather than the initial memory formation.

> *My understanding is that the Amari-Hopfield network should be evolving in continuous time and not be binary. But there were no time constants mentioned, and the equations were not provided, and it seems that the elements of X were binary units, rather than analog. This should be clarified.*

Thank you for the comment.

Although there are models with continuous firing rates and continuous time (Ramsauer et al., 2021), the original Amari-Hopfield model uses binary neurons operating in discrete time steps. As we answered the comments (5) and (6) from Reviewer 1, we considered only a discretely timestepped environment for which the timescale is arbitrary. At each environmental state where the current contextual state is selected, it typically takes about ten iterations for the conversion of the Amari-Hopfield network.

In the text, we added the following text.

“For simplicity, the environment is defined in discrete time, and agents move through environmental states characterized by distinct external stimuli.”

> *Figure 3 is aimed at replicating the lap cell finding of Sun et al, 2020. In panel E, a comparison is made between the data and the model. Are the cells in the model the entire population of H neurons (state and transition), or just a subset? Does the absence of the "ghosts" (the weaker off diagonal responses seen in the experimental data) imply that the network is not encoding that it is in the same location, but a different lap? Why is there not any true sequentiality (i.e., why do all H units go on at once)?*

Thank you for your insightful comments. Throughout this study, we used 300 neurons for the Sequence composer (H); however, for simplicity, we constrained the model such that only a single H neuron was active at each time point. As a result, most other neurons remained silent. Accordingly, in Fig. 3E, we display only neurons with firing activity, and silent neurons are not shown.

As you correctly inferred, hippocampal neurons in our model encode lap identity rather than the same physical location across laps. This design choice reflects our focus on hippocampal neurons representing contextual states, rather than place-coding neurons, as only the former contributes directly to contextual behavior in our framework. As shown in Fig. 3E, hippocampal neurons exhibit clear sequential activity with “episode-like” representations corresponding to individual laps. Nevertheless, we believe that incorporating a mixture of context-coding neurons and place-coding neurons is an important direction for future work, as illustrated in Fig. S3.

We revised the caption of Fig. 3E as follows.

“E, The comparison of (Left) lap cells in the hippocampus in the 4-lap task (Sun et al., 2020) and (Right) our results of active neurons in the H module.”

> *Typo "but also makeS predictions".*

Thank you for pointing this out. We revised it correctly.

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